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Sheaf cohomology

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Grau en Matemàtiques

Introduction

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Part I

Category theory

Chapter 1

Categories, morphisms and duality

1.1 Categories

Definition 1.1.1 (Category)**.** A **category** C consists of:

- a collection $\mathrm{Ob}(\mathsf{C})$ of **objects**.
- for each pair of objects $X, Y \in \mathrm{Ob}(\mathsf{C})$ a collection $\mathrm{Hom}_{\mathsf{C}}(X, Y)$ of **morphisms from** X **to** Y.
- for each object $X \in \mathrm{Ob}(\mathsf{C})$ a morphisms $\mathbb{1}_X \in \mathrm{Hom}_{\mathsf{C}}(X, X)$ called the **identity** morphism.
- for each pair of morphisms $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ and $g \in \mathrm{Hom}_{\mathsf{C}}(Y, Z)$ a morphism $g \circ f \in \mathrm{Hom}_{\mathsf{C}}(X, Z)$ called their **composite**.

satisfying the following axioms:

- Composition is associative: for each quadruple $X, Y, Z, W \in \mathrm{Ob}(\mathsf{C})$ of objects, if $f \in \mathrm{Hom}_{\mathsf{C}}(Y, Z)$, $g \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ and $h \in \mathrm{Hom}_{\mathsf{C}}(W, X)$ then
$$(f \circ g) \circ h = f \circ (g \circ h)$$
- Composition satiesfies the unit laws: for each pair of objects $X, Y \in \mathrm{Ob}(\mathsf{C})$ if $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ then
$$\mathbb{1}_Y \circ f = f = f \circ \mathbb{1}_X$$

⌟

Notation. To indicate that X is an object of a category C we write $X \in \mathsf{C}$ instead of $X \in \mathrm{Ob}(\mathsf{C})$.

To indicate that $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ is a morphisms from X to Y we write $f : X \to Y$ or $X \xrightarrow{f} Y$.

Example 1.1.1.

- The category Set has the collection of all sets as $\mathrm{Ob}(\mathsf{Set})$ and, for any pair of sets X, Y, $\mathrm{Hom}_{\mathsf{Set}}(X, Y)$ is the set of all applications $f : X \to Y$.
- The category Grp has the collection of all groups as $\mathrm{Ob}(\mathsf{Grp})$ and, for any pair of groups G, H $\mathrm{Hom}_{\mathsf{Grp}}(G, H)$ is the set of all group homomorphisms $f : G \to H$.

- The category Top has the collection of all topological spaces as $\mathrm{Ob}(\mathsf{Top})$ and, for any pair of topological spaces X, Y $\mathrm{Hom}_{\mathsf{Top}}(X, Y)$ is the set of all continuous functions $f : X \to Y$.

$\Diamond$

Definition 1.1.2 (Locally small category)**.** A category C is **locally small** if for every pair of objects $X, Y \in \mathsf{C}$ the collection $\mathrm{Hom}(X, Y)$ is a set. ⌟

Definition 1.1.3 (Small category)**.** A locally small category C is **small** if its collection of objects $\mathrm{Ob}(\mathsf{C})$ is a set. ⌟

1.2 Types of morphisms

Definition 1.2.1. ⌟

Definition 1.2.2 (Isomorphism)**.** An **isomorphism** in a category is a morphism $f : X \to Y$ for which exists another morphism $g : Y \to X$ such that:

$$g \circ f = \mathbb{1}_X \quad \text{and} \quad f \circ g = \mathbb{1}_Y$$

The morphism g is called the **inverse** of f, often denoted f^{-1}. ⌟

Example 1.2.1.

- In Set, an isomorphism is a bijection.
- In Grp, an isomorphism is a bijective group homomorphism.
- In Top, an isomorphism is a homeomorphism.

$\Diamond$

Proposition 1.2.1. *Let $f : X \to Y$, $g : Y \to Z$ be two isomorphisms in a category C. Then:*

1. *The inverse f^{-1} of f is unique.*
2. *$g \circ f$ is an isomorphism.*
3. *If $h \circ k$ is an isomorphism and either g or h is an isomorphism then so is the other.*

1.3 Duality

Definition 1.3.1 (Opposite category)**.** Let C be a category. The **opposite category** C^{op} consists of:

- The same objects as C: $\mathrm{Ob}(\mathsf{C}^{\mathrm{op}}) = \mathrm{Ob}(\mathsf{C})$.
- For each pair of objects $X, Y \in \mathsf{C}^{\mathrm{op}}$: $\mathrm{Hom}_{\mathsf{C}^{\mathrm{op}}}(X, Y) = \mathrm{Hom}_{\mathsf{C}}(Y, X)$.

⌟

Example 1.3.1. Consider a preordered set $(P, \leq)$ viewed as a small category where a unique morphism $x \to y$ exists if and only if $x \leq y$.

The opposite category P^{op} corresponds to the same set P equipped with the reversed order $\geq$. That is, a morphism $x \to y$ exists in P^{op} if and only if $y \leq x$ in P. $\Diamond$

Chapter 2

Functors and natural transformations

2.1 Functors

Definition 2.1.1 (Functor)**.** ⌟

2.2 Natural transformations

2.3 Representable functors

Chapter 3

Limits and colimits

3.1 Diagrams, limits and colimits

3.2 Initial and terminal objects

3.3 Products and coproducts

3.4 Directed and inverse limits

Chapter 4

More on functors

4.1 Adjunctions

4.2 Functor categories

Part II

Homological algebra

Chapter 5

Additive and abelian categories

5.1 Additive categories

Definition 5.1.1 (Additive category)**.** A category A is **additive** if it verifies:

A.1) There exists a zero object $0 \in \mathsf{A}$.

A.2) A has finite categorical products.

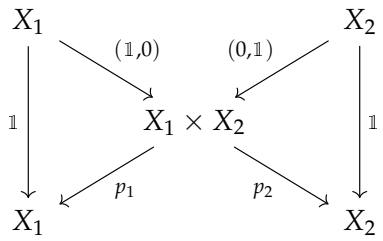
A.3) For every pair of objects $X, Y \in \mathsf{A}$ the set $\mathrm{Hom}_{\mathsf{A}}(X, Y)$ is endowed with an abelian group structure and the composition maps are bilinear with respect to the group operation.

⌟

Lemma 5.1.1. *Finite products and coproducts coincide in an additive category.*

Proof. Consider $X_1, X_2 \in \mathsf{A}$. We only need to check that $X_1 \times X_2 \cong X_1 \amalg X_2$ since the case for a finite collection of elements follows directly by induction. To do this, we will see that $X_1 \times X_2$ satisfies the universal property of the coproduct.

Consider the following diagram



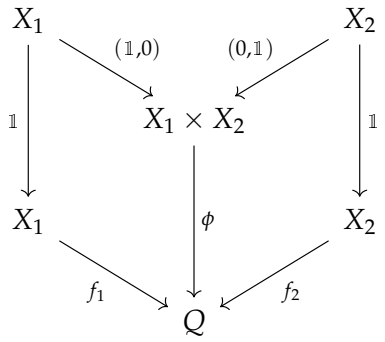
Note that for any pair of morphisms $(\alpha_1, \beta_1), (\alpha_2, \beta_2) : R \to X_1 \times X_2$ we have that

$$(\alpha_1, \beta_1) + (\alpha_2, \beta_2) = (\alpha_1 + \alpha_2, \beta_1, \beta_2)$$

Therefore if p_1, p_2 are two projections on $X_1 \times X_2$ whose sum is the identity we have that

$$(\mathbb{1}, 0) \circ p_1 + (0, \mathbb{1}) \circ p_2 = \mathbb{1}_{X_1 \times X_2}$$

Consider now two morphisms $f_i : X_i \to Q$. Then its sum $\phi = f_1 \circ p_1 + f_2 \circ p_2$ produces the following diagram



Furthermore, this is unique. Indeed, consider $\phi' : X_1 \times X_2 \to Q$ another morphism filling the diagram. Then

$$\begin{aligned}
\phi &= \phi \circ \mathbb{1}_{X_1 \times X_2} \\
&= \phi \circ ((\mathbb{1}, 0) \circ p_1 + (0, \mathbb{1}) \circ p_2) \\
&= \phi \circ (id, 0) \circ p_1 + \phi \circ (0, \mathbb{1}) \circ p_2 \\
&= f_1 \circ p_1 + f_2 \circ p_2 \\
&= \phi' \circ (\mathbb{1}, 0) + \phi' \circ (0, \mathbb{1}) \\
&= \phi' \circ ((\mathbb{1}, 0) \circ p_1 + (0, \mathbb{1}) \circ p_2) \\
&= \phi' \circ \mathbb{1}_{X_1 \times X_2} \\
&= \phi'
\end{aligned}$$

Therefore $X_1 \times X_2$ verifies the universal property of $X_1 \amalg X_2$. □

Notation. If $\{X_i\}_{i=1}^n$ is a finite set of objects in an additive category we will denote its product (and therefore by Lemma 5.1.1 its coproduct) by $\bigoplus_{i=1}^n X_i$ and call it its **direct sum**.

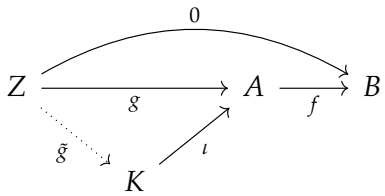
Lemma 5.1.2. *A morphism $\varphi : A \to B$ in an additive category is a monomorphism if and only if for every morphism $\xi : Z \to A$*

$$\varphi \circ \xi = 0 \implies \xi = 0$$

Dually φ is an epimorphism if and only if for every morphism $\xi : B \to Z$

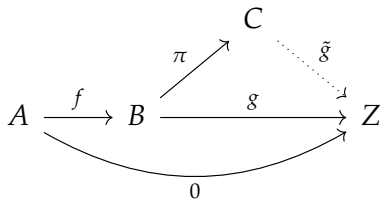
$$\xi \circ \varphi = 0 \implies \xi = 0$$

Definition 5.1.2 (Kernel)**.** Let $f : A \to B$ be a morphism in an additive category. A morphism $\iota : K \to A$ is a **kernel** of f if $f \circ \iota = 0$ and for every morphism $g : Z \to A$ such that $f \circ g = 0$ there exists a unique $\tilde{g} : Z \to K$ making the following diagram commute



⌟

Definition 5.1.3 (Cokernel)**.** Let $f : A \to B$ be a morphism in an additive category. A morphism $\pi : B \to C$ is a **cokernel** of f if $\pi \circ f = 0$ and for every morphism $g : B \to Z$ such that $\pi \circ g = 0$ there exists a unique $\tilde{g} : C \to Z$ making the following diagram commute

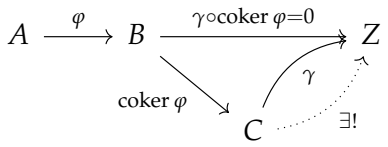


⌟

Lemma 5.1.3. *In an additive category kernels are monomorphisms and cokernels are epimorphisms.*

Proof. We will only show that cokernels are epimorphisms. The statement about kernels follows analogously.

Let $\varphi : A \to B$ be a morphism and let $\operatorname{coker} \varphi : B \to C$ be its cokernel. Consider another morphism $\gamma : C \to Z$ such that $\gamma \circ \operatorname{coker} \varphi = 0$. Since $(\gamma \circ \operatorname{coker} \varphi) \circ \varphi = 0$ by definition of the cokernel we have



Therefore since $\gamma \circ \operatorname{coker} \varphi = 0 = 0 \circ \operatorname{coker} \varphi$ uniqueness forces $\gamma = 0$ and thus by Lemma 5.1.2 φ is an epimorphism. □

Lemma 5.1.4. *Let $\varphi : A \to B$ be a morphism in an additive category. Then φ is a monomorphism if and only if $0 \to A$ is its kernel and φ is an epimorphism if and only if $B \to 0$ is its cokernel.*

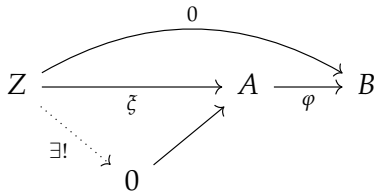
Proof. We will do the kernel one.

$\Longrightarrow$ Suppose φ is a monomorphism.

If $\xi : Z \to A$ is a morphism such that $\varphi \circ \xi = 0$ then by Lemma 5.1.2 $\xi = 0$. In particular ξ factors uniquely through $0 \to A$, i.e: $0 \to A$ is the kernel of φ.

$\Longleftarrow$ Suppose now that $0 \to A$ is the kernel of φ.

Consider a morphism $\xi : Z \to A$ such that $\varphi \circ \xi = 0$. By the kernel properties we have the following diagram



Therefore by uniqueness $\xi = 0$, making φ a monomorphism by Lemma 5.1.2.

□

5.1.1 Additive functors

Definition 5.1.4 (Additive functor)**.** Let A, B be two additive categories. A functor $F : \mathsf{A} \to \mathsf{B}$ is **additive** if it preserves the abelian group structure of all Hom-sets. ⌟

5.2 Abelian categories

Definition 5.2.1 (Abelian category)**.** An additive category A is **abelian** if it verifies

A.4) Kernels and cokernels exist in A.

A.5) Every monomorphism is the kernel of some morphism and every epimorphism is the cokernel of some morphism.

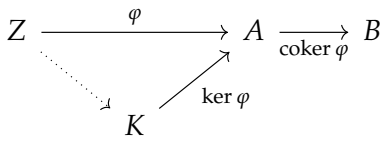
⌟

Lemma 5.2.1. *In an abelian category* A *every kernel is the kernel of its cokernel and every cokernel is the cokernel of its kernel. In formulas*

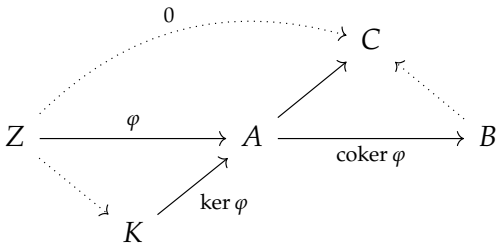
$$\ker(\operatorname{coker}\varphi) = \ker\varphi \qquad \operatorname{coker}(\ker\varphi) = \operatorname{coker}\varphi$$

Proof. Lets check cokernels this time.

Let $\operatorname{coker}\varphi : A \to B$ be the cokernel of a morphism $\varphi : Z \to A$. Since A is abelian φ has a kernel $\ker\varphi : K \to A$. Note that $Z \to A \to B$ is 0, therefore by the kernel universal property we have



Consider a morphism $A \to C$ such that $(K \to A \to C) = 0$. Now using the cokernel universal property we obtain

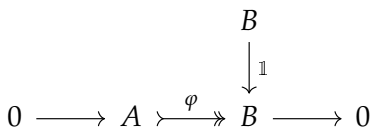


Therefore $\operatorname{coker}\varphi$ satisfies the universal property of the cokernel of $\ker\varphi : K \to A$. □

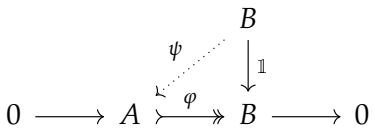
Lemma 5.2.2. *Let $\varphi : A \to B$ be a morphism in an abelian category* A*. If φ is both a monomorphism and epimorphism then φ is an isomorphism.*

Proof. By Lemma 5.1.4 the kernel of φ is $0 \to A$ and its cokernel is $B \to 0$. Furthermore by Lemma 5.2.1 φ is the cokernel of $0 \to A$ and the kernel of $B \to 0$.

Consider the following diagram

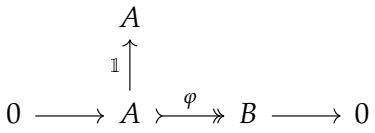


Since $(B \to B \to 0) = 0$ and φ is the kernel of $B \to 0$ we have

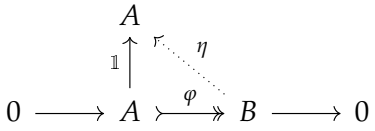


Therefore since $\varphi\psi = \mathbb{1}$ we have shown that φ has a right-inverse.

Consider now the diagram



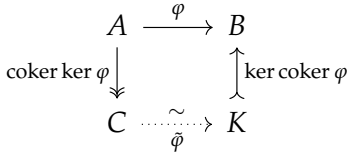
Since $(0 \to A \to A) = 0$ and φ is the cokernel of $0 \to A$ we have



Therefore since $\eta\varphi = \mathbb{1}$ we have shown that φ has a left-inverse. □

The factorization property

Theorem 5.2.1 (Factorization property)**.** *Every morphism $\varphi : A \to B$ in an abelian category* A *can be decomposed as*



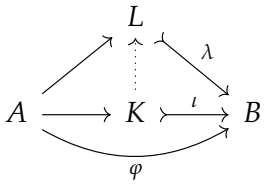
Before proving this statement, we need to consider some auxiliary lemmas.

Lemma 5.2.3. *Let $\varphi : A \to B$ be a morphism in an abelian category and let $\iota : K \to B$ be the kernel of its cokernel. Then*

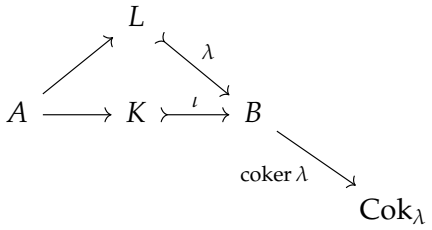
- *ι is a monomorphism.*
- *φ factors through ι.*
- *ι is initial with these properties.*

Proof.

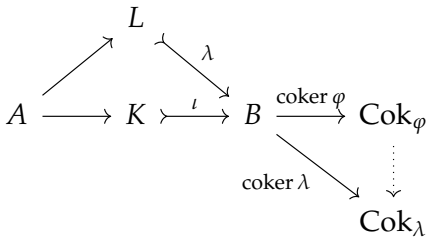
- By Lemma 5.1.3 ι is a monomorphism.
- Since $A \to B \to \operatorname{coker}\varphi$ is the zero morphism by the universal property of kernels there is an induced morphism $K \to A$.
- We want to see that if $\lambda : L \rightarrowtail B$ is a monomorphism that factors through φ then ι factors uniquely through λ



Consider coker λ



Since φ factors through λ we have that $A \to B \to \mathrm{Cok}_\lambda = 0$. Therefore by the universal property of the cokernel we have an induced map



Therefore since $K \to \mathrm{Cok}_\varphi = 0$ we have that $K \to B \xrightarrow{\operatorname{coker}\lambda} \mathrm{Cok}_\lambda = 0$.

Finally since by Lemma 5.2.1 $\lambda : L \to B$ is the kernel of its cokernel there exists a unique morphism $K \to L$ making the diagram commute.

□

Definition 5.2.2. Let $\varphi : A \to B$ be a morphism in an abelian category. The **image** of φ is

$$\operatorname{im}\varphi = \ker\operatorname{coker}\varphi$$

and the **coimage** of φ is

$$\operatorname{coim}\varphi = \operatorname{coker}\ker\varphi$$

⌟

Notation. If $\varphi : A \rightarrowtail B$ is a monomorphism we denote by ${}^{B}\!/_{A}$ the image of coker φ.

In this way the target of coim φ is ${}^{B}\!/_{\ker\varphi}$.

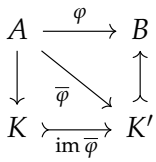
Lemma 5.2.4. *Let $\varphi : A \to B$ be a morphism in an abelian category with $\operatorname{im}\varphi : K \to B$ and $\operatorname{coim}\varphi : A \to C$. Then the induced morphisms $A \to K$ and $B \to C$ are, respectively, an epimorphism and a monomorphism.*

Proof. We will see that $\overline{\varphi} : A \to K$ is a epimorphism. The other statement follows analogously.

Consider the image of $\overline{\varphi}$

$$K' \xrightarrow{\operatorname{im}\overline{\varphi}} K \xrightarrow{\operatorname{coker}\overline{\varphi}} \mathrm{Cok}$$

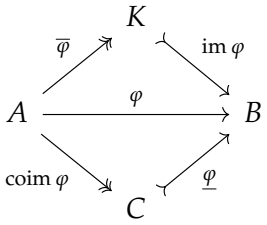
Since by Lemma 5.2.1 every epimorphism is the cokernel of its kernel we have that $\operatorname{coker} \overline{\varphi} = \operatorname{coker} \operatorname{im} \overline{\varphi}$. Moreover since $K' \rightarrowtail K$ is a monomorphism through which $\overline{\varphi}$ factors we have



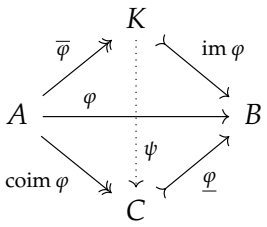
Since $\operatorname{im} \varphi$ is initial among this kind of morphisms by Lemma 5.2.3 necessarily $\operatorname{im} \overline{\varphi}$ is an isomorphism.

Therefore $0 = \operatorname{coker} \operatorname{im} \overline{\varphi} = \operatorname{coker} \overline{\varphi}$, proving that $\overline{\varphi}$ is an epimorphism by Lemma 5.1.4. □

To summarize, we have the following situation



Proof of Theorem 5.2.1. By the universal property of $\operatorname{im} \varphi$ and $\operatorname{coim} \varphi$ there exists a unique morphism $\psi : K \to C$ making the above diagram commute



Since $\varphi \circ \psi = \operatorname{im} \varphi$ is a monomorphism so is ψ. Since $\psi \circ \overline{\varphi} = \operatorname{coim} \varphi$ is a epimorphism so is ψ. Therefore by Lemma 5.2.2 ψ is an isomorphism.

Taking $\tilde{\varphi} = \psi^{-1} : C \to K$ we are done. □

5.3 Exactness

Definition 5.3.1 (Exact sequence)**.** An sequence of morphisms

$$X \xrightarrow{f} Y \xrightarrow{g} Z$$

in an abelian category A is **exact** at Y if $f \circ g = 0$ and $\operatorname{im} f = \ker g$. ⌟

Definition 5.3.2 (Short exact sequence)**.** An sequence of morphisms

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

in an abelian-category A is **exact** is called a **short exact sequence**.

Exact complexes of the form $0 \to X \to Y \to Z$ (resp. $X \to Y \to Z \to 0$) are called left (resp. right) short exact sequences. ⌟

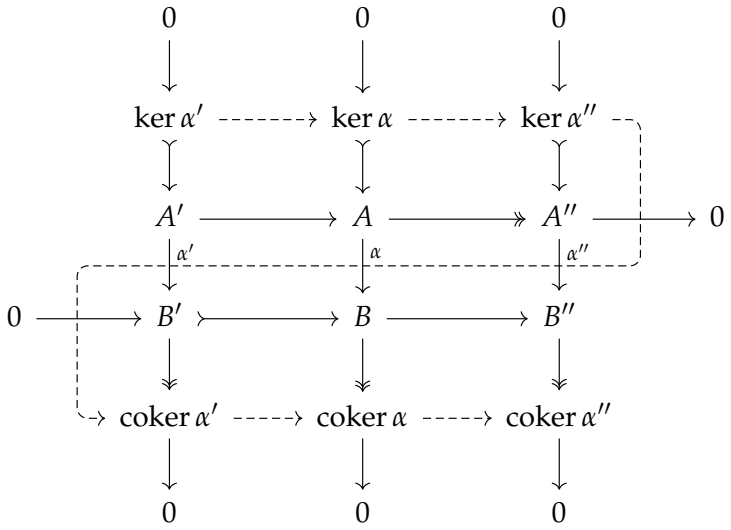
Remark 5.3.1. For any morphism $f : X \to Y$ in an A we get short exact sequences

$$0 \longrightarrow \ker f \longrightarrow X \longrightarrow \operatorname{im} f \longrightarrow 0$$

$$0 \longrightarrow \operatorname{im} f \longrightarrow Y \longrightarrow \operatorname{coker} f \longrightarrow 0$$

Therefore by Lemma 5.1.4 the sequence $0 \to X \xrightarrow{f} Y$ (resp. $X \xrightarrow{f} Y \to 0$) is exact if and only if f is a monomorphism (resp. epimorphism). ▽

Theorem 5.3.1 (Snake lemma)**.** *In any abelian category, any diagram of the form*

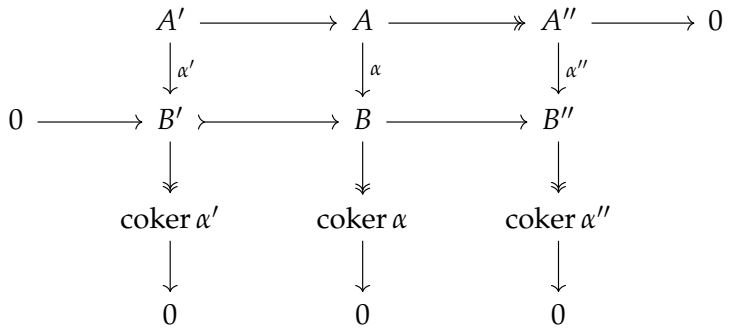


with exact rows and columns can be completed by dashed arrows making this sequence exact.

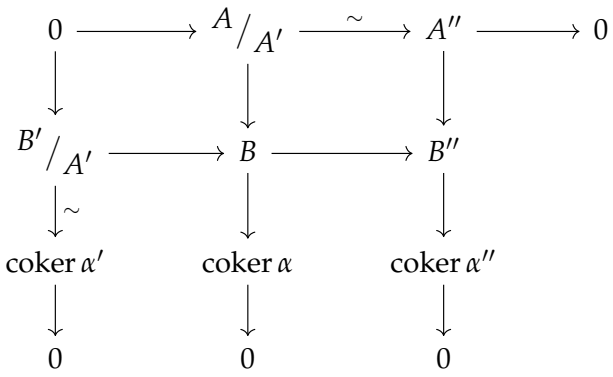
Proof. We will give a purely *arrow-theoretic* proof following [Wis].

We will be using Lemma 5.1.3 and Lemma 5.2.2 throughout the proof without explicit mention.

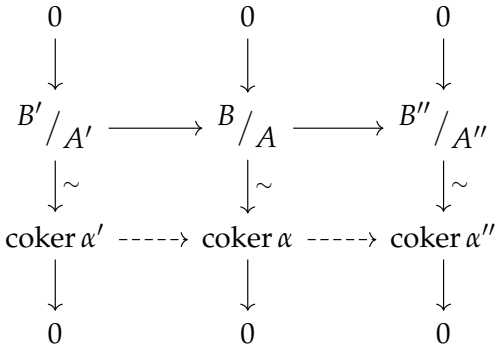
Consider first the diagram



Take first cokernels by A'



And then take cokernels by A/A'



This produces an exact sequence $\operatorname{coker}\alpha' \to \operatorname{coker}\alpha \to \operatorname{coker}\alpha''$.

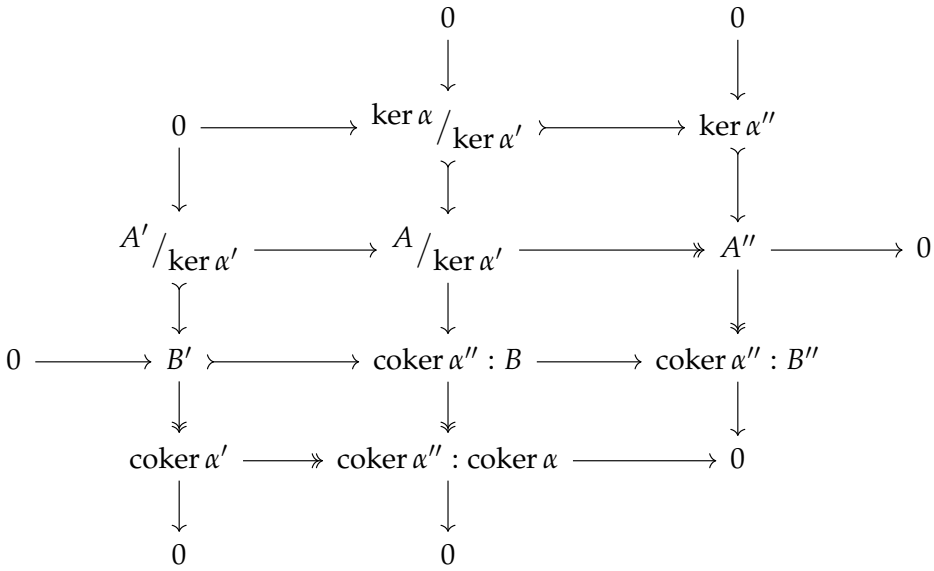
Taking kernels instead we get an exact sequence $\ker\alpha' \to \ker\alpha \to \ker\alpha''$.

It remains to produce a map $\delta : \ker\alpha'' \to \operatorname{coker}\alpha'$ and show that it induces an isomorphism

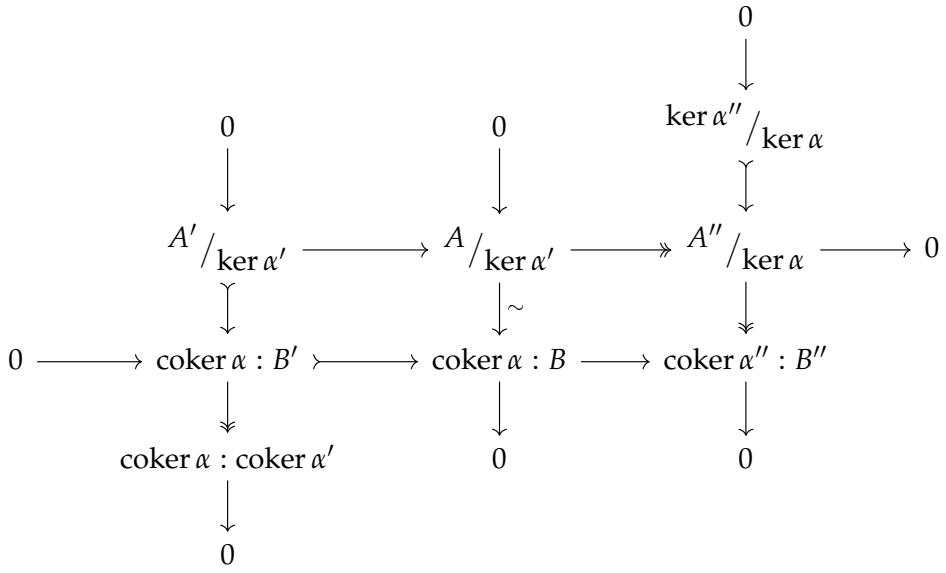
$$\operatorname{coker}(\ker\alpha \to \ker\alpha'') \cong \ker(\operatorname{coker}\alpha \to \operatorname{coker}\alpha')$$

To simplify the notation, we will denote $\ker(\operatorname{coker}\alpha \to \operatorname{coker}\alpha'')$ by $\operatorname{coker}\alpha : \operatorname{coker}\alpha''$.

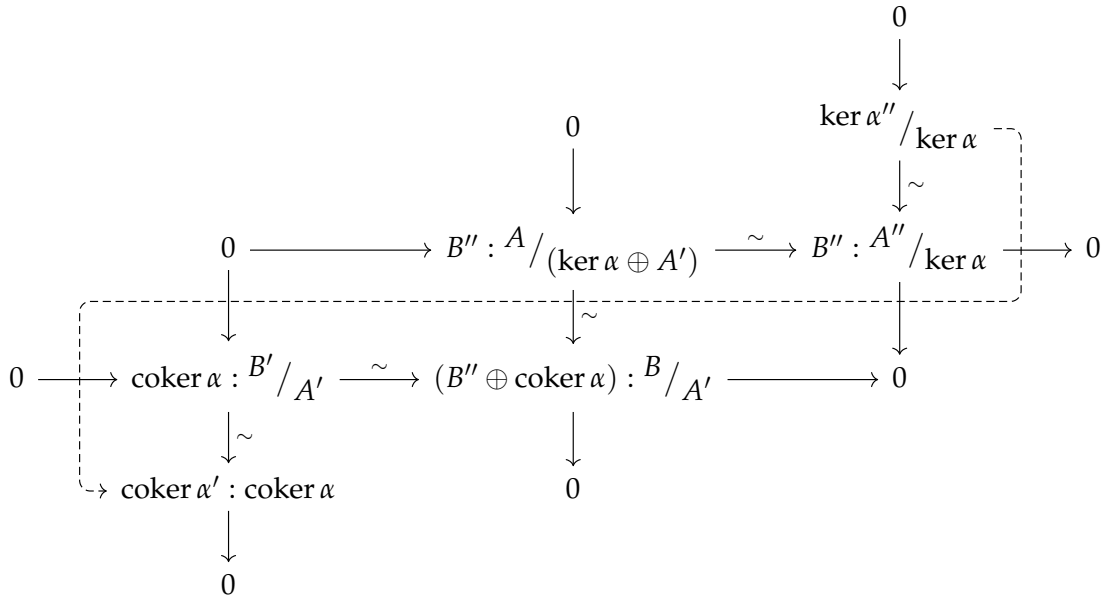
First, take cokernels by $\ker\alpha'$ and kernels into $\operatorname{coker}\alpha''$ in the full diagram



Now, take cokernels by $\ker\alpha/\ker\alpha'$ and kernels into $\operatorname{coker}\alpha : \operatorname{coker}\alpha''$



Finally, taking cokernels by $A'/\ker\alpha'$ and kernels into $\operatorname{coker}\alpha'' : B''$ we get



□

5.3.1 Exact functors

Definition 5.3.3 (Exact functor)**.** Let A, B be two abelian categories.

An additive functor $F : \mathsf{A} \to \mathsf{B}$ is called left (resp. right) **exact** if for any short exact sequence in A

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z$$

(resp. $X \to Y \to Z \to 0$) the squence

$$0 \longrightarrow F(X) \longrightarrow F(Y) \longrightarrow F(Z)$$

(resp. $F(X) \to F(Y) \to F(Z) \to 0$) is exact.

If F is both left and right exact, then F is called **exact**. ⌟

Example 5.3.1. The functors $\mathrm{Hom}_{\mathsf{A}}(X, -)$ and $\mathrm{Hom}_{\mathsf{A}}(-, X)$ are both left-exact functors from A to Ab. ◊

Proof. We will check the statement about $\mathrm{Hom}_{\mathsf{A}}(X, -)$, the other follows analogously.

Consider a short left-exact sequence

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C$$

in A. We want to show that

$$0 \longrightarrow \mathrm{Hom}(X, A) \xrightarrow{f_*} \mathrm{Hom}(X, B) \xrightarrow{g_*} \mathrm{Hom}(X, C)$$

is exact in Ab.

By Remark 5.3.1, to show exacteness at $\mathrm{Hom}(X, A)$ we need to check that f_* is a monomorphism in Ab, i.e: f_* is injective.

Let $\alpha \in \mathrm{Hom}(X, A)$ such that $f_*(\alpha) = f \circ \alpha = 0$. Since $0 \to A \xrightarrow{f} B$ is exact by **??** f is a monomorphism. Therefore $f \circ \alpha = 0 \iff \alpha = 0$, proving that f_* is injective.

To see exactness at $\mathrm{Hom}(X, B)$ we are going to check that $\mathrm{im}\, f_* = \ker g_*$:

- $\subseteq$ Let $\phi = f_*(\alpha)$ for some $\alpha \in \mathrm{Hom}(X, A)$. Applying g_* we get $g_*(\phi) = (g \circ f) \circ \alpha$, therefore by exactness of the original sequence $\phi \in \ker g_*$.
- $\supseteq$ Let $\beta \in \mathrm{Hom}(X, B)$ such that $g_*(\beta) = 0$. Since the original sequence is exact $\ker g = f$, therefore by the universal property of the kernel there exists $\alpha : X \to A$ such that $f \circ \alpha = \beta$.

□

The following is [Sch, Lemma 5.2.2].

Lemma 5.3.1. *Let $F : \mathsf{A} \to \mathsf{B}$ be an additive functor.*

1. *The following are equivalent*
   (a) *F is left exact.*
   (b) *F commutes with kernels: for any morphism $f : X \to Y$, $F(\ker f) \xrightarrow{\sim} \ker F(f)$.*
   (c) *F commutes with finite limits.*
   (d) *For any short exact sequence $0 \to X \to Y \to Z \to 0$ in A the sequence $0 \to F(X) \to F(Y) \to F(Z)$ is exact in B.*
2. *The following are equivalent*
   (a) *F is exact.*
   (b) *For any exact sequence $X \to Y \to Z$ in A the sequence $F(X) \to F(Y) \to F(Z)$ is exact in B.*
   (c) *F commutes with finite limits and finite colimits.*

Chapter 6

Complexes and cohomology

6.1 Complexes and cohomology

Throughout this chapter we will be working over a fixed abelian category A.

Definition 6.1.1 (Complex)**.** A sequence of objects and morphisms

$$\cdots \xrightarrow[d_{i+2}]{} C_{i+1} \xrightarrow[d_{i+1}]{} C_i \xrightarrow[d_i]{} C_{i-1} \xrightarrow[d_{i-1}]{} \cdots$$

in A is a **chain complex** if for every $i \in \mathbb{Z} \quad d_i \circ d_{i+1} = 0$. We denote a chain complex as $(C_\bullet, d_\bullet)$ or simply as $C_\bullet$.

By reversing the order, we say that a sequence

$$\cdots \xrightarrow{d^{i-2}} C^{i-1} \xrightarrow{d^{i-1}} C^i \xrightarrow{d^i} C^{i+1} \xrightarrow{d^{i+1}} \cdots$$

is a **cochain complex** if for every $i \in \mathbb{Z} \quad d^i \circ d^{i-1} = 0$. We denote a cochain complex as $(C^\bullet, d^\bullet)$ or simply as $C^\bullet$. ⌟

Remark 6.1.1. In purelly algebraic terms, chains are cochains are *equivalent*. We will adopt the cochain notation, but all proofs and definitions translate to the chain notation. Therefore from now onwards we will be refering to *cochains* as *chains* or *sequences*. ▽

Definition 6.1.2 (Morphism of complexes)**.** Let $C^\bullet, D^\bullet$ be two complexes in A. A **morphism of complexes** $f : C^\bullet \to D^\bullet$ is a collection of morphisms $\{f^i : C^i \to D^i\}_{i\in\mathbb{Z}}$ making, for every $i \in \mathbb{Z}$, the following diagram commute

$$\begin{array}{ccc} C^i & \xrightarrow{d^i_C} & C^{i+1} \\ {\scriptstyle f^i}\downarrow & & \downarrow{\scriptstyle f^{i+1}} \\ D^i & \xrightarrow{d^i_D} & D^{i+1} \end{array}$$

⌟

Definition 6.1.3. Given any abelian category A we can construct a **complex category** $\mathsf{Ch}(\mathsf{A})$ by taking $\mathrm{Ob}_{\mathsf{Ch}(\mathsf{A})}$ to be the collection of all complexes with values in A and, for every $C^\bullet, D^\bullet \in \mathrm{Ob}_{\mathsf{Ch}(\mathsf{A})}$, $\mathrm{Hom}_{\mathsf{Ch}(\mathsf{A})}(C^\bullet, D^\bullet)$ to be the collection of all morphisms of complexes between $C^\bullet$ and $D^\bullet$. ⌟

Remark 6.1.2. A morphism of complexes $f : C^\bullet \to D^\bullet$ is a monomorphism (resp. epimorphism) if for every $i \in \mathbb{Z}$ the morphism $f^i : C^i \to D^i$ is a monomorphism (resp. epimorphism) in A. ▽

Theorem 6.1.1. *If* A *is an abelian-category, so is its category of complexes* $\mathsf{Ch}(\mathsf{A})$.

Proof. By taking addition *elementwise* and $0^\bullet$ as the zero object we can make $\mathsf{Ch}(\mathsf{A})$ into an additive category. Therefore we only need to prove that $\mathsf{Ch}(\mathsf{A})$ verifies the axioms of an abelian category.

Let $f : A^\bullet \to B^\bullet$ be a morphism of complexes.

Since A is abelian, for every $i \in \mathbb{Z}$ the map $f^i : A^i \to B^i$ admits a kernel $\ker f^i : K^i \rightarrowtail A^i$ and a cokernel $\operatorname{coker} f^i : B^i \twoheadrightarrow C^i$.

Consider the diagram

$$\begin{array}{ccc}
K^i & & K^{i+1} \\
{\scriptstyle \ker f^i}\downarrow & & \downarrow{\scriptstyle \ker f^{i+1}} \\
A^i & \xrightarrow{d_C^i} & A^{i+1} \\
{\scriptstyle f^i}\downarrow & & \downarrow{\scriptstyle f^{i+1}} \\
B^i & \xrightarrow{d_D^i} & B^{i+1} \\
{\scriptstyle \operatorname{coker} f^i}\downarrow & & \downarrow{\scriptstyle \operatorname{coker} f^{i+1}} \\
C^i & & C^{i+1}
\end{array}$$

Since the inner square is commutative we get unique morphisms $K^i \to K^{i+1}$ and $C^i \to C^{i+1}$ making the whole diagram commute, obtaining chain complexes $K^\bullet$ and $C^\bullet$ and morphisms of chains $\ker f^\bullet, \operatorname{coker} f^\bullet$. It is then easy to check that $\ker f^\bullet$ and $\operatorname{coker} f^\bullet$ verify the corresponding universal properties.

On the other hand, by Remark 6.1.2 the other axiom is easily verified *element by element*. □

Definition 6.1.4 (Bounded complexes)**.** Let $C^\bullet$ be a complex. We say that $C^\bullet$ is

- **bounded below** if there exists $k \in \mathbb{Z}$ such that $\forall i \leq k \quad C^i = 0$.
- **bounded above** if there exists $k \in \mathbb{Z}$ such that $\forall i \geq k \quad C^i = 0$.
- **bounded** if it is both bounded below and above.

⌟

Notation. We denote by $\mathsf{Ch}^{-,+,b}(\mathsf{A}) \hookrightarrow \mathsf{Ch}(\mathsf{A})$ the full subcategory of the chain complex category consisting of bounded below, bounded above and bounded complexes respectively.

We also have variants $\mathsf{Ch}^{\geq 0,\leq 0}(\mathsf{A}) \hookrightarrow \mathsf{Ch}(\mathsf{A})$ of complexes with $\forall i \geq 0$ (resp $\forall i \leq 0$) $C^i = 0$.

By Theorem 6.1.1 all of this categories are abelian.

Remark 6.1.3. For all integers $i \in \mathbb{Z}$ there exists a fully faithfull functor

$$\iota_r : \mathsf{A} \to \mathsf{Ch}(\mathsf{A})$$

sending an object $A \in \mathsf{A}$ to a complex

$$\cdots \longrightarrow 0 \xrightarrow{d^{r-1}} A \xrightarrow{d^r} 0 \longrightarrow \cdots$$

and a morphism $\varphi : A \to B$ in A to a morphism of complexes

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & 0 & \xrightarrow{d^{r-1}} & A & \xrightarrow{d^r} & 0 & \longrightarrow & \cdots \\ & & \downarrow & & \downarrow{\scriptstyle \varphi} & & \downarrow & & \\ \cdots & \longrightarrow & 0 & \xrightarrow{d^{r-1}} & B & \xrightarrow{d^r} & 0 & \longrightarrow & \cdots \end{array}$$

In this way, we can identify A as its *image* via $\iota = \iota_0$ in $\mathsf{Ch}(\mathsf{A})$. ▽

Definition 6.1.5 (Shift functor)**.** For any integer $i \in \mathbb{Z}$ we define the **shift** functor

$$\begin{aligned} \mathsf{Ch}(\mathsf{A}) &\to \mathsf{Ch}(\mathsf{A}) \\ M^\bullet &\mapsto M[i]^\bullet \end{aligned}$$

by $M[i]^k = M^{k+i}$ and $d^k_{M[i]^\bullet} = (-1)^i d^{k+i}_{M^\bullet}$. ⌟

Definition 6.1.6 (Exact complex)**.** A complex $C^\bullet$ is **exact** at position i if the sequence

$$C^{i-1} \xrightarrow{d^{i-1}} C^i \xrightarrow{d^i} C^{i+1}$$

is exact.

If the complex is exact at every position, we say that it is **exact**. ⌟

6.2 Cohomology

Definition 6.2.1 (Cohomology)**.** The ***i*-th cohomology** of a complex $C^\bullet$ is

$$\mathrm{H}^i(C^\bullet) = {\ker d^i}\big/{\operatorname{im} d^{i-1}}$$

⌟

Remark 6.2.1. The definition of cohomology should be parsed as follows: let $I \to C^i$ be $\operatorname{im} d^{i-1}$ and $K \to C^i$ be $\ker d^i$. Since $C^\bullet$ is a complex $\operatorname{im} d^{i-1}$ factors through $\ker d^i$, giving a monomorphism $I \rightarrowtail K$. Then $\mathrm{H}^i(C^\bullet) = \operatorname{coker}(I \rightarrowtail K)$. ▽

Proposition 6.2.1. *For every integer i, the assignement $\mathrm{H}^i : M^\bullet \mapsto H^i(M^\bullet)$ taking a complex $M^\bullet$ to its i-th cohomology group is a functor $\mathsf{Ch}(\mathsf{A}) \to \mathsf{A}$.*

Proof. We want to show that for all complexes $M^\bullet, N^\bullet \in \mathsf{Ch}(\mathsf{A})$ and $i \in \mathbb{Z}$, H^i induces in a natural way a homomorphism of abelian groups

$$\operatorname{Hom}_{\mathsf{Ch}(\mathsf{A})}(M^\bullet, N^\bullet) \to \operatorname{Hom}_{\mathsf{A}}(\mathrm{H}^i(M^\bullet), \mathrm{H}^i(N^\bullet))$$

Let $\alpha^\bullet : M^\bullet \to N^\bullet$ be a morphism of complexes. Let's consider the relevant part of the commutative diagram defining $\alpha^\bullet$:

$$\begin{array}{ccccc} M^{i-1} & \xrightarrow{d^{i-1}_{M^\bullet}} & M^i & \xrightarrow{d^i_{M^\bullet}} & M^{i+1} \\ {\scriptstyle \alpha^{i-1}}\downarrow & & {\scriptstyle \alpha^i}\downarrow & & {\scriptstyle \alpha^{i+1}}\downarrow \\ N^{i-1} & \xrightarrow{d^{i-1}_{N^\bullet}} & N^i & \xrightarrow{d^i_{N^\bullet}} & N^{i+1} \end{array}$$

By the commutativity of the right square we have that $d^i_{N^\bullet} \circ \alpha^i = \alpha^{i+1} \circ d^i_{M^\bullet}$ is 0 on $\ker d^i$. Therefore the restriction of α^i to $\ker d^i_{M^\bullet}$ factors through $\ker d^i_{N^\bullet}$. Composing with the projection we get a morphism

$$\ker d^i_{M^\bullet} \to \mathrm{H}^i(N^\bullet)$$

Since this morphism is 0 when restricted to $\operatorname{im} d_{M^\bullet}^{i-1}$ by the commutativity of the left square we get that α^i induces a morphism

$$\mathrm{H}^i(M^\bullet) \to \mathrm{H}^i(N^\bullet)$$

in A as we wanted. □

Remark 6.2.2. By Remark 6.1.3 we can see cohomology as a functor $\mathrm{H}^i : \mathsf{Ch}(\mathsf{A}) \to \mathsf{Ch}(\mathsf{A})$ using ι_i. Note that for this particular complex $M^\bullet$ we have that

$$\mathrm{H}^\bullet(\mathrm{H}^\bullet(M^\bullet)) = \mathrm{H}^\bullet(M^\bullet)$$

▽

Theorem 6.2.1 (Long Exact Sequence in Cohomology)**.** *Every short exact sequence*

$$0 \longrightarrow A^\bullet \xrightarrow{f^\bullet} B^\bullet \xrightarrow{g^\bullet} C^\bullet \longrightarrow 0$$

in $\mathsf{Ch}(\mathsf{A})$ *induces a long exact sequence in cohomology*

$$\begin{array}{l} \cdots \longrightarrow \mathrm{H}^{i-1}(B) \longrightarrow \mathrm{H}^{i-1}(C) \\ \quad \hookrightarrow \mathrm{H}^i(A) \longrightarrow \mathrm{H}^i(B) \longrightarrow \mathrm{H}^i(C) \\ \quad\quad \hookrightarrow \mathrm{H}^{i+1}(A) \longrightarrow \mathrm{H}^{i+1}(B) \longrightarrow \cdots \end{array}$$

Proof. For every integer i, applying the snake lemma to the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A^i & \xrightarrow{f^i} & B^i & \xrightarrow{g^i} & C^i & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle d_A^i} & & \downarrow{\scriptstyle d_B^i} & & \downarrow{\scriptstyle d_C^i} & & \\ 0 & \longrightarrow & A^{i+1} & \xrightarrow{f^{i+1}} & B^{i+1} & \xrightarrow{g^{i+1}} & C^{i+1} & \longrightarrow & 0 \end{array}$$

we get an exact sequence

$$\operatorname{coker} d_A^i \longrightarrow \operatorname{coker} d_B^i \longrightarrow \operatorname{coker} d_C^i \longrightarrow 0$$

Applying now the snake lemma to the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A^{i+1} & \xrightarrow{f^{i+1}} & B^{i+1} & \xrightarrow{g^{i+1}} & C^{i+1} & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle d_A^{i+1}} & & \downarrow{\scriptstyle d_B^{i+1}} & & \downarrow{\scriptstyle d_C^{i+1}} & & \\ 0 & \longrightarrow & A^{i+2} & \xrightarrow{f^{i+2}} & B^{i+2} & \xrightarrow{g^{i+2}} & C^{i+2} & \longrightarrow & 0 \end{array}$$

we get an exact sequence

$$0 \longrightarrow \ker d_A^{i+1} \longrightarrow \ker d_B^{i+1} \longrightarrow \ker d_C^{i+1}$$

Since $d^{i+1} \circ d^i = 0$ we have the following diagram

$$\begin{array}{ccccccc} & & \operatorname{coker} d_A^i & \longrightarrow & \operatorname{coker} d_B^i & \longrightarrow & \operatorname{coker} d_C^i & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \overline{d}_A^i} & & \downarrow{\scriptstyle \overline{d}_B^i} & & \downarrow{\scriptstyle \overline{d}_C^i} & & \\ 0 & \longrightarrow & \ker d_A^{i+1} & \longrightarrow & \ker d_B^{i+1} & \longrightarrow & \ker d_C^{i+1} & & \end{array}$$

Applying the snake lemma to this diagram we get an exact sequence

$$\begin{array}{ccccccc} \ker \overline{d}_A^i & \longrightarrow & \ker \overline{d}_B^i & \longrightarrow & \ker \overline{d}_A^i & \to & \\ & & & \overset{\delta^i}{\searrow} & & & \\ \operatorname{coker} \overline{d}_A^i & \longrightarrow & \operatorname{coker} \overline{d}_B^i & \longrightarrow & \operatorname{coker} \overline{d}_C^i & & \end{array}$$

It only remains to prove that $\ker \overline{d}_A^i \cong \mathrm{H}^i(A)$ and $\operatorname{coker} \overline{d}_A^i \cong \mathrm{H}^{i+1}(A)$.

Consider the diagram

$$\begin{array}{ccccccccc} & & 0 & & 0 & & 0 & & \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \operatorname{im} d_A^{i-1} & = & \operatorname{im} d_A^{i-1} & \longrightarrow & 0 & & \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \ker d_A^i & \longrightarrow & A^i & \longrightarrow & \operatorname{im} d_A^i & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathrm{H}^i(A) & \longrightarrow & {}^{A^i}\!/_{\operatorname{im} d_A^{i-1}} & \xrightarrow{\psi} & \operatorname{im} d_A^i & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ & & 0 & & 0 & & 0 & & \end{array}$$

Since the first two rows are exact by the nine lemma we have a short exact sequence

$$0 \longrightarrow \mathrm{H}^i(A) \longrightarrow {}^{A^i}\!/_{\operatorname{im} d_A^{i-1}} \xrightarrow{\psi} \operatorname{im} d_A^i \longrightarrow 0$$

Therefore since $\overline{d}_A^i : {}^{A^i}\!/_{\operatorname{im} d_A^{i-1}} \xrightarrow{\psi} \operatorname{im} d_A^i \xrightarrow{\iota} \ker d_A^{i+1}$ we have

$$\ker(\overline{d}_A^i) = \ker \psi \cong \mathrm{H}^i(A^\bullet)$$

On the other hand, note that $\operatorname{coker} \overline{d}_A^i = \operatorname{coker}(\iota \circ \psi) = \operatorname{coker} \iota$ and

$$\operatorname{coker} \iota = {}^{\ker d_A^{i+1}}\!/_{\operatorname{im} \iota} = {}^{\ker d_A^{i+1}}\!/_{\operatorname{im} d_A^i} \cong \mathrm{H}^{i+1}(A)$$

□

Triangles

Consider a short exact sequence of complexes

$$0 \longrightarrow A^\bullet \longrightarrow B^\bullet \longrightarrow C^\bullet \longrightarrow 0$$

Note that we can map $C^\bullet$ to $A^\bullet$ by sending all the complex $C^\bullet$ to 0 *within* $A^\bullet$

$$C^\bullet \xrightarrow{0} A^\bullet$$

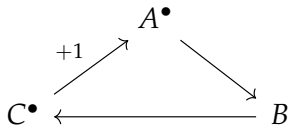
In this way, the exactness of the original sequence is equivalent to the exactness at the center of

$$C^\bullet \xrightarrow{0} A^\bullet \longrightarrow B^\bullet$$

$$A^\bullet \longrightarrow B^\bullet \longrightarrow C^\bullet$$

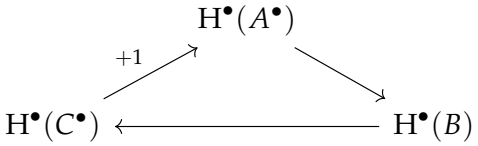
$$B^\bullet \longrightarrow C^\bullet \xrightarrow{0} A^\bullet$$

We can represent this using the *exact triangle*



The arrow marked by $+1$ in this case refers to the zero morphism. We use $+1$ to record that we can view this morphism from $C^\bullet$ to the shift $A[1]^\bullet$. Therefore, the triangle is shorthand for the diagram

Using Theorem 6.2.1 we get another *exact triangle*



However this time the $+1$ morphism is, generally, non-zero.

Chapter 7

Cones and homotopy

7.1 The mapping cone of a morphism

Definition 7.1.1 (Mapping cone)**.** Let $\alpha^\bullet : L^\bullet \to M^\bullet$ be a morphism of complexes. We define the **mapping cone** $\mathrm{Cone}(\alpha)^\bullet$ as the complex with objects

$$\forall i \in \mathbb{Z} \quad \mathrm{Cone}(\alpha)^i = L[1]^i \oplus M^i = L^{i+1} \oplus M^i$$

and differentials

$$\forall i \in \mathbb{Z} \quad d^i_{\mathrm{Cone}(\alpha)^\bullet}(\ell, m) = (-d^{i+1}_{L^\bullet}(\ell), \alpha^{i+1}(\ell) + d^i_{M^\bullet}(m))$$

Pictorially

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & L^{i+1} & \xrightarrow{-d^{i+1}_L} & L^{i+1} & \longrightarrow & \cdots \\ & \searrow & \oplus & \searrow^{\alpha^{i+1}} & \oplus & \searrow & \\ \cdots & \longrightarrow & M^i & \xrightarrow[d^i_M]{} & M^{i+1} & \longrightarrow & \cdots \end{array}$$

⌟

Remark 7.1.1. We have morphisms of complexes $M^\bullet \to \mathrm{Cone}(\alpha)^\bullet$ and $\mathrm{Cone}(\alpha)^\bullet \to L[1]^\bullet$ induced by the morphisms $M^i \to L[1]^i \oplus M^i \to L[1]^i$.

Furthermore, since the we have short exact sequences

$$0 \longrightarrow M^i \longrightarrow L[1]^i \oplus M^i \longrightarrow L[1]^i \longrightarrow 0$$

the sequence of complexes

$$0 \longrightarrow M^\bullet \longrightarrow L[1]^\bullet \oplus M^\bullet \longrightarrow L[1]^\bullet \longrightarrow 0$$

is exact. ▽

Proposition 7.1.1. *There exists an exact triangle*

$$\begin{array}{ccc} & \mathrm{H}^\bullet(M^\bullet) & \\ {\scriptstyle +1}\nearrow & & \searrow \\ \mathrm{H}^\bullet(L[1]^\bullet) & \xleftarrow{\quad\delta\quad} & \mathrm{H}^\bullet(\mathrm{Cone}(\alpha)^\bullet) \end{array}$$

where the connecting morphism δ is the morphism induced by $\alpha^\bullet$ in cohomology.

Proof. The existence of the triangle comes directly from Theorem 6.2.1, so we only need to check that δ agrees with the morphism induced by $\alpha^\bullet$.

By chasing the diagram

$$\begin{array}{ccccccccc}
 & & \vdots & & \vdots & & \vdots & & \\
 & & \uparrow & & \uparrow & & \uparrow & & \\
0 & \longrightarrow & M^i & \longrightarrow & L^{i+1} \oplus M^i & \longrightarrow & L^{i+1} & \longrightarrow & 0 \\
 & & \uparrow & & \uparrow & & \uparrow & & \\
0 & \longrightarrow & M^{i-1} & \longrightarrow & L^i \oplus M^{i-1} & \longrightarrow & L^i & \longrightarrow & 0 \\
 & & \uparrow & & \uparrow & & \uparrow & & \\
 & & \vdots & & \vdots & & \vdots & &
\end{array}$$

with a class $\mathrm{H}^{i-1}(L[1]^\bullet) = \mathrm{H}^i(L^\bullet)$ represented by $\ell \in L^i$ (such that $d^i_{L^\bullet}(\ell) = 0$) we find

$$\begin{array}{ccc}
\alpha^i(\ell) & \longmapsto & (0, \alpha^i(\ell)) \\
 & & \uparrow \\
 & & (\ell, 0) \longmapsto \ell
\end{array}$$

which verifies the claim. □

Corollary 7.1.1. *Let $\alpha^\bullet : L^\bullet \to M^\bullet$ be a morphism of complexes. Then the induced morphism $\mathrm{H}^\bullet(L^\bullet) \to \mathrm{H}^\bullet(M^\bullet)$ is an isomorphism if and only if the mapping cone $\mathrm{Cone}(\alpha)^\bullet$ is an exact complex.*

Proof. Using Proposition 7.1.1, consider the following segment of the long exact sequence

$$\begin{array}{l}
\cdots \longrightarrow \mathrm{H}^{i-1}(\mathrm{Cone}(\alpha)^\bullet) \longrightarrow \mathrm{H}^i(L^\bullet) \xrightarrow{\alpha^*} \\
\quad \mathrm{H}^i(M^\bullet) \longrightarrow \mathrm{H}^i(\mathrm{Cone}(\alpha)^\bullet) \longrightarrow \mathrm{H}^{i+1}(L^\bullet) \xrightarrow{\alpha^*} \\
\quad \mathrm{H}^{i+1}(M^\bullet) \longrightarrow \mathrm{H}^{i+1}(\mathrm{Cone}(\alpha)^\bullet) \longrightarrow \cdots
\end{array}$$

$\Longrightarrow$ Suppose α^* is an isomorphism. Since $\mathsf{Ch}(\mathsf{A})$ is abelian by Lemma 5.2.2 α^* is both a monomorphism and an epimorphism. Consider

$$\cdots \xrightarrow{\alpha^*} \mathrm{H}^i(M^\bullet) \xrightarrow{f} \mathrm{H}^i(\mathrm{Cone}(\alpha)^\bullet) \xrightarrow{g} \mathrm{H}^{i+1}(L^\bullet) \xrightarrow{\alpha^*} \cdots$$

Using that the first α^* is a surjective and exactness we have that

$$\ker f = \operatorname{im} \alpha^* = \mathrm{H}^i(M^\bullet)$$

i.e: f is the zero map.

Using that the last α^* is injective and exactness we have that

$$\ker g = \ker \alpha^* = \{0\}$$

i.e: g is the zero map.

Therefore we have that $\mathrm{H}^i \operatorname{Cone}(\alpha)^\bullet = 0$, meaning that the complex is exact.

$\Longleftarrow$ Suppose $\operatorname{Cone}(\alpha)^\bullet$ is an exact complex. Therefore since its cohomology vanishes at every index the sequence becomes

$$\cdots \longrightarrow 0 \longrightarrow \mathrm{H}^i(L^\bullet) \xrightarrow{\alpha^*} \mathrm{H}^i(M^\bullet) \longrightarrow 0 \longrightarrow \cdots$$

Therefore by exactness we have that α^* is an isomorphism.

□

7.2 Quasi-isomorphisms

Definition 7.2.1 (Quasi-isomorphism)**.** A morphism of complexes $\alpha^\bullet$ is a **quasi-isomorphism** (abbreviated q.i.s) if it induces an isomorphism in cohomology. ⌟

Remark 7.2.1. By Corollary 7.1.1, $\alpha^\bullet$ is a quasi-isomorphism if and only it its mapping cone $\operatorname{Cone}(\alpha)^\bullet$ is an exact complex. ▽

Example 7.2.1 (Quasi-isomorphism are not isomorphism)**.** In $\mathsf{Ch}(\mathsf{Ab})$, consider the morphism of complexes

$$\begin{array}{ccccccccccc}
\cdots & \longrightarrow & 0 & \longrightarrow & \mathbb{Z} & \xrightarrow{\cdot 2} & \mathbb{Z} & \longrightarrow & 0 & \longrightarrow & \cdots \\
 & & \downarrow & & \downarrow & & \downarrow{\scriptstyle \pi} & & \downarrow & & \\
\cdots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \mathbb{Z}/_{2\mathbb{Z}} & \longrightarrow & 0 & \longrightarrow & \cdots
\end{array}$$

where π is the obvious projection is a quasi-isomorphism. However, it is not an isomorphism because there are no (non-trivial) group homomorphisms $\mathbb{Z}/_{2\mathbb{Z}} \to \mathbb{Z}$. ◊

Lemma 7.2.1. *Let* A *be an additive category and let* $F : \mathsf{Ch}(\mathsf{A}) \to \mathsf{D}$ *be an additive functor such that, for every quasi-isomorphism* $\rho^\bullet$, $F(\rho^\bullet)$ *is an isomorphism. Let* $M^\bullet$ *be an exact complex in* $\mathsf{Ch}(\mathsf{A})$.

- *The complex* $F(M^\bullet)$ *is a zero object in* D.
- *Let* $\alpha^\bullet : L^\bullet \to N^\bullet$ *be a morphism of complexes in* $\mathsf{Ch}(\mathsf{A})$ *that factors through an exact complex*

$$\begin{array}{ccc}
L^\bullet & \xrightarrow{\alpha^\bullet} & N^\bullet \\
 & \searrow \quad \nearrow & \\
 & M^\bullet &
\end{array}$$

  Then $F(\alpha^\bullet)$ *is the zero morphism.*

Proof. Since $M^\bullet$ is exact, the zero morphism $M^\bullet \to M^\bullet$ is a quasi-isomorphism. Therefore we have

$$F(M^\bullet) \xrightarrow{F(0)} F(M^\bullet) \xrightarrow{F(0)^{-1}} F(M^\bullet), \qquad \mathbb{1}_{F(M^\bullet)}$$

Since F is additive $F(0) = 0$, making $\mathbb{1}_{F(M^\bullet)} = 0$.

For the other part, applying F to the diagram we get

$$\begin{array}{ccc} F(L^\bullet) & \xrightarrow{F(\alpha^\bullet)} & F(N^\bullet) \\ & \searrow \quad \nearrow & \\ & F(M^\bullet) & \end{array}$$

Therefore since $F(M^\bullet)$ is a zero object in D we have that $F(\alpha^\bullet) = 0$. □

7.3 Homotopy

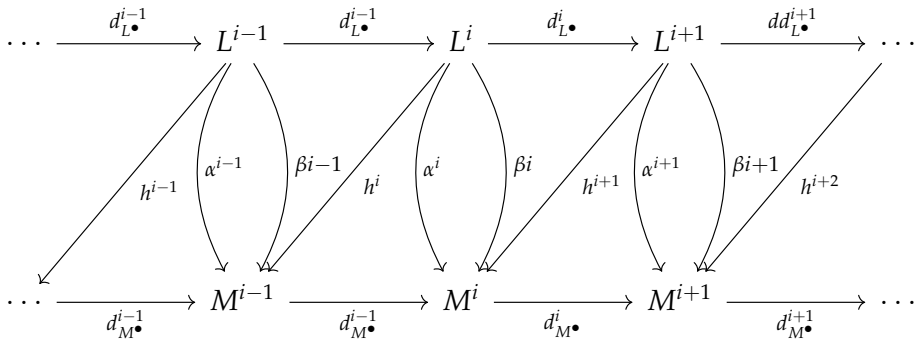
Definition 7.3.1. A **homotopy** h between two morphisms of complexes $\alpha^\bullet, \beta^\bullet : L^\bullet \to M^\bullet$ is a collection of morphisms

$$h^i : L^i \to M^{i-1}$$

such that

$$\forall i \in \mathbb{Z} \quad \beta^i - \alpha^i = d^{i-1}_{M^\bullet} \circ h^i + h^{i+1} \circ d^i_{L^\bullet}$$

Pictorially, we have the following **non-commutative** diagram



If there is an homotopy between $\alpha^\bullet$ and $\beta^\bullet$ we say that $\alpha^\bullet$ is **homotopic** to $\beta^\bullet$ and denote it by $\alpha^\bullet \sim \beta^\bullet$. ⌟

Definition 7.3.2. A morphism $\alpha^\bullet : L^\bullet \to M^\bullet$ is a **homotopy equivalence** if there exists a morphism $\beta^\bullet : M^\bullet \to L^\bullet$ such that $\alpha^\bullet \circ \beta^\bullet \sim \mathbb{1}_{M^\bullet}$ and $\beta^\bullet \circ \alpha^\bullet \sim \mathbb{1}_{N^\bullet}$. In this case, the complexes $L^\bullet$ and $M^\bullet$ are said to be **homotopy equivalent**. ⌟

Proposition 7.3.1. *If $\alpha^\bullet, \beta^\bullet : L^\bullet \to M^\bullet$ are homotopic morphisms of complexes then $\alpha^\bullet, \beta^\bullet$ induce the same morphisms on cohomology* $\mathrm{H}^\bullet(L^\bullet) \to \mathrm{H}^\bullet(M^\bullet)$.

Proof. Let $\overline{\ell} \in \mathrm{H}^i(L^\bullet)$. We have that $\overline{\ell}$ is represented by an element $\ell \in \ker d^i_{L^\bullet}$, and its images in $\mathrm{H}^i(M^\bullet)$ under α^*, β^* are represented by $\alpha^i(\ell), \beta^i(\ell)$.

Since $\alpha^\bullet, \beta^\bullet$ are homotopic there exist morphisms $h^i : L^i \to M^{i-1}$ such that

$$\beta^i(\ell) - \alpha^i(\ell) = d^{i-1}_{M^\bullet}(h^i(\ell)) + h^{i+1}(\cancelto{0}{d^i_{L^\bullet}(\ell)})$$

Therefore

$$\beta^i(\ell) - \alpha^i(\ell) \in \operatorname{im} d^i_{M^\bullet}$$

making $\beta^i(\ell) - \alpha^i(\ell)$ vanish in $\mathrm{H}^i(M^\bullet)$ as needed. □

Remark 7.3.1. If $\alpha^\bullet$ is homotopic to the identity by Corollary 7.1.1 the mapping cone of $\alpha^\bullet$ must be exact. ▽

Corollary 7.3.1. *Homotopy equivalent complexes have isomorphic cohomology.*

Proof. If $\alpha^\bullet : L^\bullet \to M^\bullet$ and $\beta^\bullet : M^\bullet \to L^\bullet$ are morphisms of complexes such that $\beta^\bullet \circ \alpha^\bullet$ and $\alpha^\bullet \circ \beta^\bullet$ are homotopic to the identity then by Proposition 7.3.1 they induce inverse morphisms in cohomology. □

Lemma 7.3.1. *Let $F : \mathsf{A} \to \mathsf{B}$ be an additive functor.*

- *If $\alpha^\bullet \sim \beta^\bullet$ are homotopic in $\mathsf{Ch}(\mathsf{A})$ then $\mathsf{Ch}(F)(\alpha^\bullet) \sim \mathsf{Ch}(F)(\beta^\bullet)$ in $\mathsf{Ch}(\mathsf{B})$.*
- *If $L^\bullet, M^\bullet$ are homotopy equivalent complexes in $\mathsf{Ch}(\mathsf{A})$ then $\mathsf{Ch}(F)(L^\bullet)$ and $\mathsf{Ch}(F)(M^\bullet)$ are homotopy equivalent complexes in $\mathsf{Ch}(\mathsf{B})$.*

Proof. Note that the second assertion follows directly from the first.

To see the first, if h is a homotopy between $\alpha^\bullet, \beta^\bullet$ then

$$\forall i \in \mathbb{Z} \quad \beta^i - \alpha^i = d^{i-1}_{M^\bullet} \circ h^i + h^{i+1} \circ d^i_{L^\bullet}$$

Since F is additive

$$\forall i \in \mathbb{Z} \quad F(\beta^i) - F(\alpha^i) = F(d^{i-1}_{M^\bullet}) \circ F(h^i) + F(h^{i+1}) \circ F(d^i_{L^\bullet})$$

showing that the collection $\{F(h^i)\}_{i\in\mathbb{Z}}$ produces a homotopy between $F(\alpha^\bullet)$ and $F(\beta^\bullet)$. □

The following is direct from the previous statements.

Theorem 7.3.1. *Let $F : \mathsf{A} \to \mathsf{B}$ be an additive between two abelian categories A, B. If $L^\bullet, M^\bullet$ are homotopy equivalent complexes in $\mathsf{Ch}(\mathsf{A})$ then the cohomology complexes*

$$\mathrm{H}^\bullet(\mathsf{Ch}(F)(L^\bullet)) \cong \mathrm{H}^\bullet(\mathsf{Ch}(F)(M^\bullet))$$

are isomorphic.

Chapter 8

Injective and projective objects

8.1 Injective and projective objects

Definition 8.1.1.

- An object $I \in \mathsf{A}$ is **injective** if all the diagrams of the form

$$\begin{array}{ccc} X & \longrightarrow & I \\ \downarrow & & \\ Z & & \end{array}$$

  where $X \rightarrowtail Z$ is a monomorphism admit an extension

$$\begin{array}{ccc} X & \longrightarrow & I \\ \downarrow & \nearrow & \\ Z & & \end{array}$$

- An object $P \in \mathsf{A}$ is **projective** if all the diagrams of the form

$$\begin{array}{ccc} & & X \\ & & \downarrow \\ P & \longrightarrow & Z \end{array}$$

  where $X \twoheadrightarrow Z$ is an epimorphism admit an extension

$$\begin{array}{ccc} & \nearrow & X \\ & & \downarrow \\ P & \longrightarrow & Z \end{array}$$

⌟

Proposition 8.1.1.

- *An object $I \in \mathsf{A}$ is injective if and only if the functor $\mathrm{Hom}_{\mathsf{A}}(-, I)$ is exact.*
- *An object $P \in \mathsf{A}$ is projective if and only if the functor $\mathrm{Hom}_{\mathsf{A}}(P, -)$ is exact.*

Proof. By Example 5.3.1 we only need to check right exactness.

We will do the injective version this time:

$\Longrightarrow$ Suppose $I \in A$ is injective and consider a short exact sequence

$$0 \longrightarrow X \xrightarrow{f} Y \xrightarrow{g} Z \longrightarrow 0$$

Using Lemma 5.3.1, we want to see that

$$0 \longrightarrow \mathrm{Hom}(Z, I) \xrightarrow{g^*} \mathrm{Hom}(Y, I) \xrightarrow{f^*} \mathrm{Hom}(X, I)$$

is exact.

Since by Example 5.3.1 this sequence is exact at $\mathrm{Hom}(Z, I)$ and $\mathrm{Hom}(Y, I)$, we only need to check exactness at $\mathrm{Hom}(X, I)$, i.e: we want to see that f^* is surjective.

Consider a morphism $\alpha : X \to I$. Since f is a monomorphism and I is injective there exists a morphism $\beta : Y \to I$ such that $\beta \circ f = f^*(\beta) = \alpha$, making f^* surjective as wanted.

$\Longleftarrow$ Suppose $\mathrm{Hom}(-, I)$ is exact and consider a monomorphism $f : A \rightarrowtail B$ in A.

We have the following short exact sequence

$$0 \longrightarrow A \xrightarrow{f} B \longrightarrow \operatorname{coker} f \longrightarrow 0$$

Applying $\mathrm{Hom}(-, I)$ to this sequence we get

$$0 \longrightarrow \mathrm{Hom}(\operatorname{coker} f, I) \longrightarrow \mathrm{Hom}(B, I) \xrightarrow{f^*} \mathrm{Hom}(A, I) \longrightarrow 0$$

which is exact since $\mathrm{Hom}(-, I)$ is exact. In particular we have that f^* is surjective.

Therefore if $\gamma : A \to I$ is any morphism there exists $\beta \in \mathrm{Hom}(B, I)$ such that $\gamma = \beta \circ f$, which is exactly the universal property of injective objects.

□

We haven't proven existence of injective or projective objects. This is because in general they may not exist.

Since we will be interesed in using them, we introduce the following types of categories.

Definition 8.1.2. Let A be an abelian category. We say that:

- A has **enough injectives** if for every object $A \in \mathsf{A}$ there exists an injective object $I \in \mathsf{A}$ and a monomorphism $A \rightarrowtail I$.
- A has **enough projectives** if for every object $A \in \mathsf{A}$ there exists a projective object $P \in \mathsf{A}$ and an epimorphism $P \twoheadrightarrow A$.

⌟

8.2 Resolutions and homotopy

Definition 8.2.1.

- An **injective resolution** of an object $A \in \mathsf{A}$ is a quasi-isomorphism $\iota(\mathsf{A}) \to Q^\bullet$, where $Q^\bullet$ is a complex in $\mathsf{Ch}^{\geq 0}(I)$.
- A **projective resolution** of an object $A \in \mathsf{A}$ is a quasi-isomorphism $P^\bullet \to \iota(\mathsf{A})$, where $P^\bullet$ is a complex in $\mathsf{Ch}^{\leq 0}(P)$.

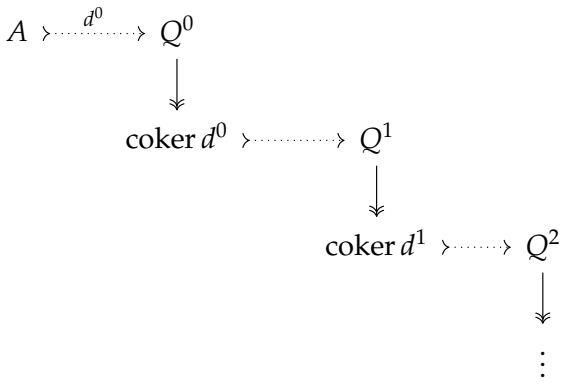
We will refer to this resolutions using $Q^\bullet, P^\bullet$, *forgetting* about the quasi-isomorphism. ⌟

Lemma 8.2.1. *Let* A *be an abelian category.*

- *If* A *has enough injectives then every object* $A \in \mathsf{A}$ *admits an injective resolution.*
- *If* A *has enough projectives then every object* $A \in \mathsf{A}$ *admits a projective resolution.*

Proof. We will do the enough injectives case.

Since A has enough injectives there exists an injective object $Q^0 \in \mathsf{A}$ and a monomorphism $d^0 : A \rightarrowtail Q^0$. Then there exists an injective object $Q^1 \in \mathsf{A}$ an a monomorphism $d^1 : \operatorname{coker} d^0 \rightarrowtail Q^0$ and so on. Pictorially



Therefore following this diagram we get an injective resolution

$$Q^\bullet = \cdots \longrightarrow 0 \longrightarrow Q^0 \longrightarrow Q^1 \longrightarrow Q^2 \longrightarrow \cdots$$

□

We will assume that A is an abelian-cateogory with enough injectives.

By *flipping* the corresponding arrows, all the results may be adapted to the projective case.

Lemma 8.2.2. *Let* A *be an object in a* A*, let* $M^\bullet$ *be a resolution of* A *and let* $Q^\bullet$ *be a complex in* $\mathsf{Ch}^{\geq 0}(I)$*. Let* $\varphi : \mathrm{H}^0(M^\bullet) = A \to \mathrm{H}^0(Q^\bullet)$ *be an arbitrary morphism. Then*

- *There exists a morphism of complexes* $\alpha : M^\bullet \to Q^\bullet$ *inducing* φ *at the level of* H^0*.*
- *Different morphisms satisfying the above are homotopy equivalent.*

Proof. We have to define $\forall i \in \mathbb{Z} \quad \alpha^i : M^i \to Q^i$. Since for $i < 0$ we have $\alpha^i = 0$ we can truncate the complexes, obtaining

$$\begin{array}{ccccccccccc}
0 & \longrightarrow & \mathrm{H}^0(M^\bullet) = A & \xrightarrow{\varepsilon} & M^0 & \xrightarrow{d^0_{M^\bullet}} & M^1 & \xrightarrow{d^1_{M^\bullet}} & M^2 & \xrightarrow{d^2_{M^\bullet}} & \cdots \\
 & & \downarrow{\scriptstyle \varphi} & & \downarrow{\scriptstyle \alpha^1} & & \downarrow{\scriptstyle \alpha^1} & & \downarrow{\scriptstyle \alpha^2} & & \\
0 & \longrightarrow & \mathrm{H}^0(Q^\bullet) & \xrightarrow[\iota]{} & Q^0 & \xrightarrow[d^0_{Q^\bullet}]{} & Q^1 & \xrightarrow[d^1_{Q^\bullet}]{} & Q^2 & \xrightarrow[d^2_{Q^\bullet}]{} & \cdots
\end{array}$$

Note that the top complex is exact.

We will construct $\alpha^\bullet$ using induction:

- Case $n = 0$. Consider the following diagram

$$\begin{array}{ccc} A & \xrightarrow{\varepsilon} & M^0 \\ \downarrow{\scriptstyle \varphi} & & \\ \mathrm{H}^0(Q^\bullet) & \xrightarrow[\iota]{} & Q^0 \end{array}$$

  Since Q^0 is injective using its universal property with the morphism $\iota \circ \varphi$ we obtain morphism $\alpha^0 : M^0 \to Q^0$ making the diagram commute.

- Inductive step. Suppose we have already construced $\alpha^0, \dots, \alpha^{n-1}$ such that

$$\forall k \in \{1, \dots, n-1\} \quad \alpha^k \circ d_{M^\bullet}^{k-1} = d_{Q^\bullet}^k \circ \alpha^{k-1}$$

  We want to construct $\alpha^n : M^n \to Q^n$.

  Consider the diagram

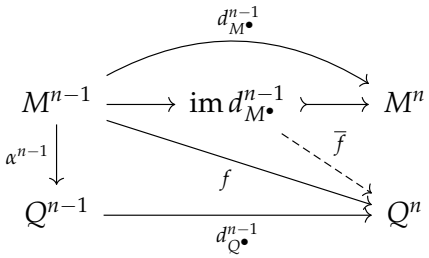
$$\begin{array}{ccc} M^{n-1} & \xrightarrow{d_{M^\bullet}^{n-1}} & M^n \\ {\scriptstyle \alpha^{n-1}}\downarrow & & \\ Q^{n-1} & \xrightarrow[d_{Q^\bullet}^{n-1}]{} & Q^n \end{array}$$

  Let $f = d_{Q^\bullet}^{n-1} \circ \alpha^{n-1} : M^{n-1} \to Q^n$. Note that

$$f \circ d_{M^\bullet}^{n-2} = d_{Q^\bullet}^{n-1} \circ \alpha^{n-1} \circ d_{M^\bullet}^{n-2} = \underbrace{d_{Q^\bullet}^{n-1} \circ d_{Q^\bullet}^{n-2}}_{0} \circ \alpha^{n-2} = 0$$

  meaning that f vanishes on $\operatorname{im} d_{M^\bullet}^{n-2} = \ker d_{M^\bullet}^{n-1}$ by exactness of $M^\bullet$.

  Therefore f induces a map $\overline{f} : \operatorname{im} d_{M^\bullet}^{n-1} \to Q^n$



  Using that Q^n is injective we obtain a morphism $\alpha^n : M^n \to Q^n$.

Falta unicitat

□

Proposition 8.2.1. *Any two injective resolutions (resp. projective) of an object A in an abelian category* A *are homotopy equivalent.*

Proposition 8.2.2. *Let A_0, A_1 be two objects in an abelian category* A*. Let $Q_i^\bullet$ be an injective resolution of $A_i, i \in \{0, 1\}$. Then every morphism $\varphi : A_0 \to A_1$ in* A *is induced by a morphism $\alpha^\bullet : I_0^\bullet \to I_1^\bullet$, uniquely determined up to homotopy.*

Part III

Sheaves

Chapter 9

Presheaves and sheaves

9.1 Presheaves

We begin by giving a general definition of a presheaf.

Definition 9.1.1 (Presheaf)**.** Let X be a topological space and let $\mathsf{Op}(X)$ be its `ex:open-set-category`. Let C be a Locally small category. The Locally small category of C-valued **presheaves** is the `ex:functor-category`

$$\mathsf{PSh}_X(\mathsf{C}) := [\mathsf{Op}(X)^{\mathrm{op}}, \mathsf{C}]$$

When the topological space is clear we will write $\mathsf{PSh}(\mathsf{C})$.

The elements of this category are Functors $\mathcal{F} : \mathsf{Op}(X)^{\mathrm{op}} \to \mathsf{C}$ and morphisms are **??**s $\phi : \mathcal{F} \to \mathcal{G}$. ⌟

Let's rewrite Definition 9.1.1 less abstractly: a C-valued Presheaf $\mathcal{F}$ is the collection of data

- For every open set $U \subseteq X$ an object $\mathcal{F}(U)$ of C called *sections of $\mathcal{F}$ over U*.
- For every inclussion of open sets $V \subseteq U$ a morphism $\mathrm{res}_{U,V} : \mathcal{F}(U) \to \mathcal{F}(V)$ in C called *restriction* such that
  - For every open set U, $\mathrm{res}_{U,U} = \mathbb{1}_U$.
  - For every inclussion of open sets $W \subseteq V \subseteq U$, $\mathrm{res}_{U,V} \circ \mathrm{res}_{V,W} = \mathrm{res}_{U,W}$.

A *morphism of Presheaves* $\phi : \mathcal{F} \to \mathcal{G}$ is a collection of morphisms $\{\phi(U) : \mathcal{F}(U) \to \mathcal{G}(U)\}_{U \in \mathsf{Op}(X)}$ in C making the following diagram commute:

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\phi(U)} & \mathcal{G}(U) \\ \downarrow{\scriptstyle \mathrm{res}_{U,V}} & & \downarrow{\scriptstyle \mathrm{res}_{U,V}} \\ \mathcal{F}(V) & \xrightarrow{\phi(V)} & \mathcal{G}(V) \end{array}$$

We are going to study now **??**s and **??**s in $\mathsf{PSh}(\mathsf{C})$.

Note that, since $\mathsf{PSh}(\mathsf{C})$ is a `def:functor-category` then if C has **??**s or **??**s of a certain shape then $\mathsf{PSh}(\mathsf{C})$ also has them and they are computed pointwise.

This means that **??**s and **??**s in $\mathsf{PSh}(\mathsf{C})$ can be computed *open set by open set*.

Proposition 9.1.1. *Let X be a topological space. Let $\mathcal{I} \to \mathsf{PSh}_X(\mathsf{C}), i \mapsto \mathcal{F}_i$ be a `def:diagram`. If C has **??**s and/or **??**s with shape $\mathcal{I}$ then $\lim_i \mathcal{F}_i$ and/or $\mathrm{colim}_i \mathcal{F}_i$ exist in $\mathsf{PSh}_X(\mathsf{C})$. Moreover*

for every open set $U \subseteq X$

$$\left(\lim_i \mathcal{F}_i\right)(U) = \lim_i \mathcal{F}_i(U) \qquad \left(\operatorname*{colim}_i \mathcal{F}_i\right)(U) = \operatorname*{colim}_i \mathcal{F}_i(U)$$

Write a section on functor categories (or quote it from somewhere) to justify this.

In particular if C is an Abelian category then Kernels and Cokernels exist and are computed *open set by open set*, making $\mathsf{PSh}(\mathsf{C})$ an Abelian category.

Remark 9.1.1. From this point onwards, we will suppose C has all **??**s and **??**s unless stated otherwise. ▽

9.1.1 Presheaves and continuous maps

Definition 9.1.2 (Direct image Presheaf)**.** Let $f : X \to Y$ be a continuous map. Let $\mathcal{F}$ be a C-valued Presheaf over X. The **direct image Presheaf** of $\mathcal{F}$ via f is the Presheaf $f_*\mathcal{F}$ defined as, for every open subset $V \subseteq Y$

$$f_*\mathcal{F}(V) = \mathcal{F}(f^{-1}(U))$$

Given $W \subseteq V \subseteq Y$ open subsets, the restriction map comes from the commutativity of the diagram

$$\begin{array}{ccc} f_*\mathcal{F}(V) & = & \mathcal{F}(f^{-1}V) \\ \vdots & & \downarrow \\ f_*\mathcal{F}(V) & = & \mathcal{F}(f^{-1}V) \end{array}$$

This construction defines a Functor $f_* : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{PSh}_Y(\mathsf{C})$. ⌟

Show that the category of presheaves over a one point space *is equal* to the evaluation category.

Example 9.1.1 (Global section)**.** Let X be a topological space. Consider the one-point topological space $\{*\}$. Let $\mathcal{F}$ be a C-valued presheaf over X. The **global section** of $\mathcal{F}$ over

This defines a Functor $\Gamma(X, -) : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{C}$ ◊

Example 9.1.2 (Skyscrapper Presheaf)**.** Let X be a topological space and $p \in X$. Suppose C has a final object $* \in \mathsf{C}$. Let S be an object in C. The **skyscrapper Presheaf** supported at S $\iota_{p,*}\underline{S}_{\text{pre}}$ is the Direct image Presheaf of the Constant Presheaf $\underline{S}_{\text{pre}}$ via the inclussion $\iota : \{p\} \hookrightarrow X$:

$$(\iota_{p,*}\underline{S}_{\text{pre}})(U) = \begin{cases} S & p \in U \\ * & p \notin U \end{cases}$$

◊

Using this we can construct a *dual* Presheaf to Definition 9.1.2.

Definition 9.1.3 (Inverse image Presheaf)**.** Let $f : X \to Y$ be a continuous map. Let $\mathcal{G}$ be a C-valued Presheaf over Y. The **inverse image Presheaf** of $\mathcal{G}$ via f is the Presheaf $f^{-1}\mathcal{G}$ defined as, for every open subset $U \subseteq X$

$$f^{-1}\mathcal{G}(U) = \operatorname*{colim}_{V \supseteq f(U)} \mathcal{G}(V)$$

where the def:categorical-colimit is taken over all the open neighbourhoods V of $f(U)$.

This defines a Functor $f^{-1} : \mathsf{PSh}_Y(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$. ⌟

Remark 9.1.2. Note that the set of open neighbourhoods of $f(U)$ is a partially ordered set via reverse inclussion. Therefore if C is a concrete category we can describe $f^{-1}\mathcal{G}(U)$ as $\{(s, V) \mid f(U) \subseteq V \in \mathsf{Op}(Y), s \in \mathcal{G}(V)\}$ modulo the relation $(s, V) \sim (s', V')$ if and only if there exists $W \subseteq V \cap V'$ open with $f(U) \subseteq W$ such that $s|_W = s'|_W$. ▽

Example 9.1.3 (Constant Presheaf)**.** Let X be a topological space. Let S be an object of C. The **constant Presheaf** associated to S $\underline{S}_{\text{pre}}$ is the Inverse image Presheaf of S via the map $\{*\} \to X$. ◊

Definition 9.1.4 (Stalk)**.** Let X be a topological space and $p \in X$. Let $\mathcal{F}$ be a C-valued Presheaf over X. The **stalk** of $\mathcal{F}$ at p $\mathcal{F}_p$ is the Inverse image Presheaf of $\mathcal{F}$ via $X \to \{p\}$:

$$\mathcal{F}_p = \operatorname*{colim}_{U \ni p} \mathcal{F}(U)$$

⌟

Proposition 9.1.2. *Let $f : X \to Y$ be a continuous map. We have an def:adjunction between the Direct image Presheaf functor and the Inverse image Presheaf $f^{-1} \dashv f_*$.*

Proof. We define the def:categorical-unit $\eta : 1 \implies f_* f^{-1}$ as the map with components given by the canonical insertion maps to the colimit

$$\mathcal{F}(W) \to \operatorname*{colim}_{U \supseteq (f \circ f^{-1})(W)} \mathcal{F}(U) = f^{-1}\mathcal{F}(f^{-1}W) = f_* f^{-1}\mathcal{F}(W)$$

Note that this maps exist because $(f \circ f^{-1})(W) \subseteq W$.

On the other hand, note that if $f(V) \subseteq U$ then $V \subseteq (f^{-1} \circ f)(V) \subseteq f^{-1}(U)$, i.e: we have a restricion map $\mathcal{G}(f^{-1}(U)) \to \mathcal{G}(V)$. Therefore by the universal property of the colimit we have a unique map

$$f^{-1} f_* \mathcal{G}(V) = \operatorname*{colim}_{U \supseteq f(V)} \mathcal{G}(f^{-1}(U)) \to \mathcal{G}(V)$$

Using this maps we obtain a def:categorical-counit $\varepsilon : f^{-1} f_* \implies 1$.

Since by construction the compositions $\eta \circ \varepsilon$ and $\varepsilon \circ \eta$ are the corresponding identity morphisms, we have proven the adjunction. □

Two important particular cases of the adjunction Proposition 9.1.2 are

Corollary 9.1.1. *Let X be a topological space and $p \in X$. Let $\mathcal{F}$ be a C-valued Presheaves over X and let A be an object in C. Then*

$$\mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\underline{A}_{\text{pre}}, \mathcal{F}) \simeq \mathrm{Hom}_{\mathsf{C}}(A, \Gamma(X, \mathcal{F}))$$

Corollary 9.1.2. *Let X be a topological space and $p \in X$. Let $\mathcal{F}$ be a C-valued Presheaves over X and let A be an object in C. Then*

$$\mathrm{Hom}_{\mathsf{C}}(\mathcal{F}_p, A) \simeq \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\mathcal{F}, \iota_{p,*}\underline{A}_{\text{pre}})$$

9.2 Sheaves

We now turn our attention to a particular class of Presheaves, following [**stacks-project**].

Let's start by considering the case $\mathsf{C} = \mathsf{Set}$:

Definition 9.2.1 (Sheaf of sets)**.** Let X be a topological space. A Presheaf of sets $\mathcal{F}$ is a **sheaf** if for every open set U and open covering $\{U_i\}_{i\in\mathcal{I}}$ of U the diagram

$$\mathcal{F}(U) \to \prod_{i\in\mathcal{I}} \mathcal{F}(U_i) \rightrightarrows \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} \mathcal{F}(U_i \cap U_j) \tag{9.2.1}$$

is an `def:equalizer`.

We denote by $\mathsf{Sh}_X(\mathsf{Set})$ (or just Sh if the context is clear) the `def:full-subcategory` of the category of presheaves of sets whose objects are sheaves of sets.

Less abstractly this means that $\mathcal{F}$ verifies:

SH.1) **Identity**: Given two sections $s, t \in \mathscr{F}(U)$ such that for every $i \in \mathcal{I}$ $\quad s|_{U_i} = t|_{U_i}$ we have that $s = t$.

SH.2) **Gluability**: Given a collection of sections $\{s_i \mid s_i \in \mathscr{F}(U_i)\}_{i\in\mathcal{I}}$ that agree on pairwise intersections ($\forall i, j \in \mathcal{I}\ \ s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}$) there exists a section $s \in \mathscr{F}(U)$ such that $\forall i \in \mathcal{I}\ \ s|_{U_i} = s_i$.

⌟

Definition 9.2.2 (Locally small category of Sheaves of sets)**.** The Locally small category $\mathsf{Sh}_X(\mathsf{Set})$ of Sheaves of sets is the `def:full-subcategory` of the category of Presheaves $\mathsf{PSh}_X(\mathsf{Set})$ whose objects are Sheaves of sets. ⌟

Let's try to make Definition 9.2.1 more general by considering another evaluation category C.

Firstly, note that if **??**s indexed by $\mathcal{I}$ and $\mathcal{I}\times\mathcal{I}$ for any indexing set $\mathcal{I}$ exist in C then the diagram (9.2.1) makes sense. Secondly, we have that (9.2.1) is an `def:equalizer` if and only if for every object $C \in \mathsf{C}$ the diagram in Set

$$\mathrm{Hom}_{\mathsf{C}}(C, \mathcal{F}(U)) \to \prod_{i\in\mathcal{I}} \mathrm{Hom}_{\mathsf{C}}(C, \mathcal{F}(U_i)) \rightrightarrows \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} \mathrm{Hom}_{\mathsf{C}}(C, \mathcal{F}(U_i \cap U_j))$$

is an `def:equalizer`.

Therefore we get

Definition 9.2.3 (Sheaf)**.** Let X be a topological space. Let C be an arbitrary category. A C-valued Presheaf $\mathcal{F}$ is a **sheaf** if for all objects $C \in \mathsf{C}$ the Presheaf of sets $\mathcal{F}_C$ defined as $\mathcal{F}_C(U) = \mathrm{Hom}_{\mathsf{C}}(C, \mathcal{F}(U))$ is a Sheaf of sets.

We denote by $\mathsf{Sh}_X(\mathsf{C})$ the `def:full-subcategory` of the category of C-valued presheaves of sets whose objects are C-valued sheaves. ⌟

Notation. From now on, to distinguish a Presheaf from a Sheaf we will denote the former by $\mathcal{F}, \mathcal{G}, \mathcal{H}, \ldots$ and the latter by $\mathscr{F}, \mathscr{G}, \mathscr{H}, \ldots$

By Definition 9.2.3 we have an inclussion functor $\iota : \mathsf{Sh}(\mathsf{C}) \hookrightarrow \mathsf{PSh}(\mathsf{C})$.

For *sufficiently nice* Locally small categorys, this Functor admits a left-adjoint $\mathsf{PSh}(\mathsf{C}) \to \mathsf{Sh}(\mathsf{C})$. Let's study first the case $\mathsf{C} = \mathsf{Set}$.

How general do we want to go?

Theorem 9.2.1 (Sheafification). *There exists a* ***sheafification*** *Functor* $(-)^{\#} : \mathsf{PSh}(\mathsf{Set}) \to \mathsf{Sh}(\mathsf{Set})$ *that is left-adjoint to the inclussion Functor* $\iota : \mathsf{Sh}(\mathsf{Set}) \to \mathsf{PSh}(\mathsf{Set})$.

Proof. Let $\mathcal{F}$ be a Presheaf of sets over X. For every open subset $U \subseteq X$ we define

$$\mathcal{F}^{\#}(U) = \left\{ f_p \in \prod_{p \in U} \mathcal{F}_p \text{ verifying } (*) \right\}$$

where $(*)$ is the property *"for all $p \in U$ there exists an open neighbourhood $V \subseteq U$ of p and a section $s \in \mathcal{F}(V)$ such that, for every $q \in V$, $s_q = f_q$"*.

Taking $\prod_{p \in U} \mathcal{F}_p \to \prod_{q \in V} \mathcal{F}_q$ as the restriction maps we see that $\mathcal{F}^{\#}$ is a Presheaf. Let's see now that $\mathcal{F}^{\#}$ is a Sheaf. To do this, consider an open subset $U \subseteq X$ with open cover $\{U_i\}_{i \in \mathcal{I}}$.

SH.1 Let $s, t \in \mathcal{F}^{\#}(U)$ be two sections such that, for all $i \in \mathcal{I}$ $s|_{U_i} = t|_{U_i}$.

Then since for every $p \in U$ there exists $i \in \mathcal{I}$ such that $p \in U_i$ we have that $s_p = (s|_{U_i})_p = (t|_{u_i})_p = t_p$. Therefore $s = t$.

SH.2 Let $\{s_i \in \mathcal{F}^{\#}(U_i)\}_{i \in \mathcal{I}}$ be a collection of sections such that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all $i, j \in \mathcal{I}$.

Consider $s \in \prod_{p \in U} \mathcal{F}_p$ defined as $s_p = (s_i)_p$ if $p \in U_i$. Note that this is well-defined by property $(*)$. To see that s verifies $(*)$ note that for every $p \in U_i$ there exists an open neighbourhood $V \subseteq U$ of p and $t \in \mathcal{F}(V)$ such that, for all $q \in V$, $t_q = (s_i)|_q = s_q$.

Let $V \subseteq U \subseteq X$ be two open sets. Consider the commutative diagram

$$\begin{array}{ccccc} \mathcal{F}(U) & \longrightarrow & \mathcal{F}^{\#}(U) & \longrightarrow & \prod_{p \in U} \mathcal{F}_p \\ \downarrow & & \downarrow & & \downarrow \\ \mathcal{F}(V) & \longrightarrow & \mathcal{F}^{\#}(V) & \longrightarrow & \prod_{q \in V} \mathcal{F}_q \end{array}$$

where the vertical arrows are the corresponding restriction maps. By chasing this diagram we get a map $\varphi : \mathcal{F} \to \mathcal{F}^{\#}$, $\varphi_U(s) = (s_p)_{p \in U}$.

It only remains to show that any morphism $\psi : \mathcal{F} \to \mathcal{G}$, where $\mathcal{G}$ is a Sheaf of sets, factors uniquely as $\mathcal{F} \xrightarrow{\varphi} \mathcal{F}^{\#} \to \mathcal{G}$, i.e: we have to construct a morphism $\theta : \mathcal{F}^{\#} \to \mathcal{G}$ such that $\psi = \theta \circ \varphi$.

Let $U \subseteq X$ be an open subset and $s \in \mathcal{F}^{\#}(U)$. For each $p \in U$ consider an open neighbourhood $V^{(p)} \subseteq U$ of p and $t^{(p)} \in \mathcal{F}(V^{(p)})$ such that $t^{(p)}_q = s_q$. Consider now $\psi_{V^{(p)}}(t^{(p)}) \in \mathcal{G}(V^{(p)})$. Note that $\{\psi_{V^{(p)}}(t^{(p)})\}_{p \in U}$ is a compatible family of section in $\mathcal{G}$, therefore by the Sheaf axioms there exists a unique section $t \in \mathcal{G}(U)$ such that $t|_{V^{(p)}} = \psi_{V^{(p)}}(s^{(p)})$. Therefore we can define $\theta_U(s) = t$.

By construction $\psi = \theta \circ \varphi$ and θ commutes with the restriction maps, making it a morphism of sheaves.

To check uniqueness, let $\theta' : \mathcal{F}^{\#} \to \mathcal{G}$ be another morphism such that $\psi = \theta' \circ \varphi$. Consider $s \in \mathcal{F}^{\#}(U)$. For each $p \in U$ choose $V^{(p)} \subseteq U$ and $s^{(p)} \in \mathcal{F}(V^{(p)})$ such that $s|_{V^{(p)}} = \varphi_{V^{(p)}}(s^{(p)})$. Then

$$\theta'(s)|_{V^{(p)}} = \theta'(s|_{V^{(p)}}) = \theta'(\varphi_{V^{(p)}}(s^{(p)})) = \psi_{V^{(p)}}(s^{(p)})$$

On the other hand $\theta(s)|_{V^{(p)}} = \psi_{V^{(p)}}(s^{(p)})$ by definition. Therefore since $\mathscr{G}$ is a Sheaf by SH.1 we have $\theta'(f) = \theta(f)$, i.e: $\theta' = \theta$. □

We can extend the construction outlined in Theorem 9.2.1 to **??**s.

Lemma 9.2.1. *Let (C, F) be an* `def:algebraic-category`*. Let $\mathcal{F}$ be a C-valued Presheaf. Then there exists a C-valued Sheaf $\mathcal{F}^{\#}$ and a morphism of presheaves $\mathcal{F} \to \mathcal{F}^{\#}$ satisfying*

1. *The map $\mathcal{F} \to \mathcal{F}^{\#}$ identifies the underlying Sheaf of sets of $\mathcal{F}^{\#}$ with the sheafification of the underlying Presheaf of sets of $\mathcal{F}$.*
2. *For any morphism of presheaves $\mathcal{F} \to \mathscr{G}$ where $\mathscr{G}$ is a C-valued Sheaf there exists a unique factorization $\mathcal{F} \to \mathcal{F}^{\#} \to \mathscr{G}$.*

Remark 9.2.1. We will assume that C is an `def:algebraic-category` in what follows. ▽

Therefore we can assign a Sheaf to any Presheaf using this construction.

Definition 9.2.4 (Sheafification)**.** Let $\mathcal{F}$ be a C-valued Presheaf. We call the Sheaf $\mathcal{F}^{\#}$ obtained by Theorem 9.2.1 the **sheafification** of $\mathcal{F}$. ⌟

Using this adjunction we can *adapt* Definition 9.1.2 and Definition 9.1.3 to the case of Sheaves

Definition 9.2.5 (Direct image Sheaf)**.** Let $f : X \to Y$ be a continuous map. Let $\mathscr{F}$ be a C-valued Sheaf over X. The **direct image Sheaf** of $\mathscr{F}$ via f is the Sheaf $f_*\mathscr{F}$ defined as, for every open subset $V \subseteq Y$

$$f_*\mathscr{F}(V) = \mathscr{F}(f^{-1}(U))$$

This construction defines a Functor $f_* : \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{Sh}_Y(\mathsf{C})$. ⌟

This means that Example 9.1.1 and Example 9.1.2 translate directly to Sheaves.

Definition 9.2.6 (Global section Functor)**.** We denote by $\Gamma(X, -) : \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{C}$ the **global section functor** described in Example 9.1.1, but now with initial category $\mathsf{Sh}_X(\mathsf{C})$. ⌟

Definition 9.2.7 (Skyscrapper Sheaf)**.** Let X be a topological space and $p \in X$. Let S be an object of C. The **skyscrapper Sheaf** supported at p $\iota_{*,p}\underline{S}$ is the Direct image Sheaf of the Constant Sheaf $\underline{S}$ via the map $\{p\} \to X$. ⌟

Rewrite this definitions

Explain why we need to sheafificate

Definition 9.2.8 (Inverse image Sheaf)**.** Let $f : X \to Y$ be a continuous map. Let $\mathscr{G}$ be a C-valued Sheaf over Y. The **inverse image Sheaf** of $\mathscr{G}$ via f is the Sheaf $f^{-1}\mathscr{G}$ defined as, for every open subset $U \subseteq X$

$$f^{-1}\mathscr{G} = (f^{-1}\mathscr{G})^{\#}$$

where the RHS of the equation is the Sheafification of the Inverse image Presheaf of $\mathscr{G}$ via f.

This defines a Functor $f^{-1} : \mathsf{PSh}_Y(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$. ⌟

Definition 9.2.9 (Constant Sheaf)**.** Let S be an object of C. The **constant Sheaf** $\underline{S}$ associated to S is the Sheafification of the Constant Presheaf $\underline{S}_{\text{pre}}$, i.e: $\underline{S} = (\underline{S}_{\text{pre}})^{\#}$. ⌟

Definition 9.2.10 (Stalk)**.** Let X be a topological space and $p \in X$. Let $\mathscr{F}$ be a C-valued Sheaf over X. The **stalk** of $\mathscr{F}$ at p $\mathscr{F}_p$ is the Inverse image Sheaf of $\mathscr{F}$ via $X \to \{p\}$:

$$\mathscr{F}_p = \operatorname*{colim}_{U \ni p} \mathscr{F}(U)$$

⌟

Using the adjointness we have the analogous of Proposition 9.1.2

Proposition 9.2.1. *Let $f : X \to Y$ be a continuous map. We have an* `def:adjunction` *between the Direct image Sheaf functor and the Inverse image Sheaf $f^{-1} \dashv f_*$.*

Proof.

$$\begin{aligned}\operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(f^{-1}\mathscr{G}, \mathscr{F}) &= \operatorname{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(f^{-1}\mathscr{G}, \mathscr{F}) \\ &= \operatorname{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\mathscr{G}, f_*\mathscr{G}) \\ &= \operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\mathscr{G}, f_*\mathscr{G})\end{aligned}$$

□

Therefore adjunctions Corollary 9.1.1 and Corollary 9.2.2 also happen when considering Sheaves.

Corollary 9.2.1. *Let X be a topological space and $p \in X$. Let $\mathscr{F}$ be a C-valued Sheaves over X and let A be an object in C. Then*

$$\operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\underline{A}, \mathscr{F}) \simeq \operatorname{Hom}_{\mathsf{C}}(A, \Gamma(X, \mathscr{F}))$$

Corollary 9.2.2. *Let X be a topological space and $p \in X$. Let $\mathscr{F}$ be a C-valued Sheaves over X and let A be an object in C. Then*

$$\operatorname{Hom}_{\mathsf{C}}(\mathscr{F}_p, A) \simeq \operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\mathscr{F}, \iota_{p,*}\underline{A})$$

For cohomology we will be interested in the case $\mathsf{C} = \mathsf{Ab}$.

Lemma 9.2.2. *Let $a : \mathsf{A} \to \mathsf{B}$ and $b : \mathsf{B} \to \mathsf{A}$ be two Functors. Suppose that*

1. *A and B are additive categories, a, b are additive functors and $b \dashv a$.*
2. *B is an Abelian category and b is left-exact.*
3. *$ba \cong \mathbb{1}_{\mathsf{A}}$.*

Then A is abelian.

Proof. Since B is an Abelian category all **??**s and **??**s exist en B.

Since b is left-adjoint it is right-exact and therefore, exact.

Let $\varphi : B_1 \to B_2$ be a morphism in B. If $K = \ker(\varphi)$ then K is the `def:equalizer` of the diagram

$$K \longrightarrow B_1 \underset{0}{\overset{\varphi}{\rightrightarrows}} B_2$$

therefore applying b we get

$$bK \longrightarrow bB_1 \underset{0}{\overset{b\varphi}{\rightrightarrows}} bB_2$$

i.e: bK is the Kernel of $b\varphi$.

Dually we get that $b\operatorname{coker}(B_1 \to B_2)$ is the Cokernel of $b\varphi$.

Therefore we have that every morphism $b\varphi$ in A admits a Kernel and a Cokernel. Moreover since $ba \cong \mathbb{1}_{\mathsf{A}}$ every morphism in A is of the form $b\varphi$, therefore all Kernels and Cokernels exist in A. Indeed for every $\psi : A_1 \to A_2$ we have that

$$\ker \psi = b\ker(a\psi) \qquad \operatorname{coker} \psi = b\operatorname{coker}(a\psi) \tag{9.2.2}$$

It only remains to show that $\operatorname{coim} \psi = \operatorname{im} \psi$. Since A has all **??**s and **??**s A has all finite **??**s and **??**s, making a left-exact. This means that a transforms Kernels into Kernels. Therefore

$$\begin{aligned}
\operatorname{coim} \psi &= \operatorname{coker}(\ker \psi \to A_1) && \text{(definition)} \\
&= b\operatorname{coker}(a(\ker \psi \to A_1)) && \text{(9.2.2)} \\
&= b\operatorname{coker}(\ker(a\psi) \to aA_1) && (a \text{ preserves kernels}) \\
&= b\operatorname{coim}(a\psi) && \text{(definition)} \\
&= b\operatorname{im}(a\psi) && (\mathsf{B} \text{ is abelian}) \\
&= b\ker(aA_2 \to \operatorname{coker}(a\psi)) && \text{(definition)} \\
&= \ker(baA_2 \to b\operatorname{coker}(a\psi)) && (b \text{ preserves kernels}) \\
&= \ker(A_2 \to b\operatorname{coker}(a\psi)) && ba \cong \mathbb{1}_{\mathsf{A}} \\
&= \ker(A_2 \to \operatorname{coker} \psi) && \text{(9.2.2)} \\
&= \operatorname{im} \psi && \text{(definition)}
\end{aligned}$$

□

Proposition 9.2.2. *Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism in $\mathsf{Sh}(\mathsf{Ab})$.*

1. *The Locally small category $\mathsf{Sh}_X(\mathsf{Ab})$ is an Abelian category.*
2. *The Kernel of φ is the same as the Kernel of φ in $\mathsf{PSh}(\mathsf{Ab})$.*
3. *φ is injective if and only if φ is injective as a map or presheaves if and only if φ is injective as a map of sheaves.*
4. *The Cokernel of φ is the Sheafification of the Cokernel of φ in $\mathsf{PSh}(\mathsf{Ab})$.*
5. *φ is surjective if and only if φ is surjective as a map of sheaves.*
6. *A complex $\mathscr{F} \to \mathscr{G} \to \mathscr{H}$ is exact at $\mathscr{G}$ if and only if for all open sets U and sections $s \in \mathscr{G}(U)$ mapping to 0 in $\mathscr{H}(U)$ there exists an open covering $\{U_i\}_{i\in\mathcal{I}}$ such that each $s|_{U_i}$ is the image of $\mathscr{F}(U_i) \to \mathscr{G}(U_i)$.*

Proof. Note that 2. $\implies$ 3., 4. $\implies$ 5 and 2. $\wedge$ 5. $\implies$ 6.

Therefore we only need to prove 1., 2. and 3., but this comes directly from Lemma 9.2.2 with $\mathsf{A} = \mathsf{Sh}_X(\mathsf{Ab})$, $\mathsf{B} = \mathsf{PSh}_X(\mathsf{Ab})$, $a : \mathsf{Sh}_X(\mathsf{Ab}) \hookrightarrow \mathsf{PSh}_X(\mathsf{Ab})$ the inclussion Functor and $b : \mathsf{PSh}_X(\mathsf{Ab}) \to \mathsf{Sh}_X(\mathsf{Ab})$ the Sheafification Functor. □

Chapter 10

Ringed spaces and $\mathscr{O}_X$-modules.

10.1 Ringed spaces

Definition 10.1.1 (Ringed space)**.** A **ringed space** is a tuple $(X, \mathscr{O}_X)$ where X is a topological space and $\mathscr{O}_X$ is a Sheaf of rings over X.

We call $\mathscr{O}_X$ the *structure sheaf* of the ringed space $(X, \mathscr{O}_X)$. Sections of $\mathscr{O}_X$ over an open subset $U \subseteq X$ are called *functions on* U and functions on X are called (global) functions. ⌟

Example 10.1.1. Let and X be a topological space. Then the $\mathbb{R}$-Algebra of continuous functions $\mathcal{C}^0_X(U) = \{f : U \to \mathbb{R} \mid f \text{ continuous}\}$ with the usual restriction maps forms a Sheaf of rings over X. Therefore $(X, \mathcal{C}^0_X)$ is a Ringed space. ◊

Example 10.1.2. If for every point $p \in X$ the Stalks $\mathscr{O}_{X,p}$ of the structure sheaf are Local rings then we call $(X, \mathscr{O}_X)$ a **locally ringed space**. ◊

Definition 10.1.2 (Morphism of Ringed spaces)**.** Let $(X, \mathscr{O}_X)$ and $(Y, \mathscr{O}_Y)$ be two Ringed spaces. A **morphism of ringed spaces** $(X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ is a pair $(f, f^\sharp)$ where $f : X \to Y$ is a continuous function (i.e: a morphism in Top) and $f^\sharp : \mathscr{O}_Y \to f_*\mathscr{O}_X$ is a morphism of sheaves of rings (i.e: a morphism in $\mathsf{Sh}_{\mathsf{Ring}}(X)$), where $f_*\mathscr{O}_X$ is the Direct image Sheaf. ⌟

Maybe it is better to define $f^\sharp$ as an f-map. **[stacks-project]**

Example 10.1.3. Let $f : X \to Y$ be a continuous function. Consider the Ringed spaces of continuous functions $(X, \mathcal{C}^0_X)$, $(Y, \mathcal{C}^0_Y)$.

Then the Morphism of Ringed spaces induced by f can be defined for every $V \subseteq Y$ open as $f^\sharp(V) : \mathcal{C}^0_Y(V) \to \mathcal{C}^0_X(f^{-1}(V))$, $f^\sharp(V)(h) = h \circ f|_{f^{-1}V}$. ◊

10.2 $\mathscr{O}_X$ modules

Definition 10.2.1 ($\mathscr{O}_X$-module)**.** Let $(X, \mathscr{O}_X)$ be a Ringed space. A **Sheaf of $\mathscr{O}_X$ modules** (or simply an $\mathscr{O}_X$-module) is a Sheaf $\mathscr{F}$ such that for every open set $U \subseteq X$ $\mathscr{F}(U)$ is an $\mathscr{O}_X(U)$-Module compatible with the restriction maps, i.e: for every $f \in \mathscr{O}_X(U)$ and $s \in \mathscr{F}(U)$: $(f \cdot s)|_V = f|_V \cdot s|_V$. ⌟

Example 10.2.1. ... ◊

Definition 10.2.2 (Morphism of $\mathscr{O}_X$-modules)**.** Let $\mathscr{F}, \mathscr{G}$ be two $\mathscr{O}_X$-modules. A morphism $\phi : \mathscr{F} \to \mathscr{G}$ is a sheaf morphism that commutes with the multiplication of the structure sheaf $\mathscr{O}_X$. ⌟

Example 10.2.2. ... ◊

Therefore for every Ringed space $(X, \mathscr{O}_X)$ the collection of $\mathscr{O}_X$-modules forms a Locally small category $\mathscr{O}_X$-Mod.

10.2.1 $\mathscr{O}_X$-Mod and sheaves of abelian groups.

Just as abelian groups are $\mathbb{Z}$-modules, Sheaves of abelian groups can be seen as $\underline{\mathbb{Z}}_X$-modules, where $\underline{\mathbb{Z}}_X$ is the `ex:constant-sheaf` with coefficients in the integers.

To show this, for every Sheaf of abelian groups $\mathscr{F}$ over X and every open set $U \subseteq X$ we have to define an action $\cdot : \underline{\mathbb{Z}}_X(U) \times \mathscr{F}(U) \to \mathscr{F}(U)$ compatible with the restriction maps.

Let $f \in \underline{\mathbb{Z}}_X(U)$ and $s \in \mathscr{F}(U)$. Since f is locally constant there exists an open cover $\{U_i\}_{i\in\mathcal{I}}$ of U such that for every $i \in \mathcal{I} \quad f|_{U_i} = n_i \in \mathbb{Z}$.

For every U_i consider $t_i = n_i \cdot s|_{U_i}$, where $n_i \cdot s|_{U_i}$ is shorthand for $s|_{U_i}$ added to itself n_i times. Note that this is well-defined because $\mathscr{F}(U_i)$ is an abelian group.

Let $U_i, U_j \in \{U_i\}_{i\in\mathcal{I}}$ such that $U_i \cap U_j = \varnothing$. For every $x \in U_i \cap U_j$ there exists an open neighbourhood $V \subseteq U_i \cap U_j$ of x were f is constant.

Therefore $n_i = n_j$ and $t_i|_{U_i\cap U_j} = t_j|_{U_i\cap U_j}$ and, applying axiom SH.2, we can define $f \cdot s = t$.

Checking the axioms in small enough open sets such that f is constant verifies that this action makes $\mathscr{F}$ an $\underline{\mathbb{Z}}_X$-module.

Do we have to actually write the check?

10.3 $\mathscr{O}_X$-Mod is abelian.

Let's check first that $\mathscr{O}_X$-Mod is a Additive category.

Lemma 10.3.1. *The category $\mathscr{O}_X$-Mod admits a zero object.*

Proof. Consider the $\mathscr{O}_X$ module that sends every open set to $\{0\}$. □

Lemma 10.3.2. *The category $\mathscr{O}_X$-Mod has all binary biproducts. We define their direct sum a $\mathscr{F} \oplus \mathscr{G} = \mathscr{F} \times \mathscr{G}$.*

Proof. Let $\mathscr{F}, \mathscr{G}$ be two $\mathscr{O}_X$-modules. □

Theorem 10.3.1. *Let $\varphi : \mathscr{F} \to \mathscr{G}$ a Morphism of $\mathscr{O}_X$-modules. Then the kernel of φ is the subsheaf of $\mathscr{F}$ with sections*

$$(\ker \varphi)(U) = \{s \in \mathscr{F}(U) \mid \varphi(s) = 0\}$$

Furthermore for every $x \in X \quad (\ker \varphi)_x = \ker \varphi_x$.

Proof. If $V \subseteq U$

$$\begin{array}{ccccccc} 0 & \longrightarrow & (\ker\varphi)(U) & \longrightarrow & \mathscr{F}(U) & \xrightarrow{\varphi(U)} & \mathscr{G}(U) \\ & & \vdots & & \downarrow{\scriptstyle \mathrm{res}_{U,V}} & & \downarrow{\scriptstyle \mathrm{res}_{U,V}} \\ 0 & \longrightarrow & (\ker\varphi)(V) & \longrightarrow & \mathscr{F}(V) & \xrightarrow{\varphi(V)} & \mathscr{G}(V) \end{array}$$

□

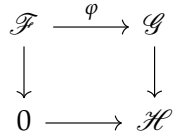
Theorem 10.3.2. *Let $\varphi : \mathscr{F} \to \mathscr{G}$ a Morphism of $\mathscr{O}_X$-modules. Then the cokernel of φ is the Sheafification of the quotient Presheaf*

$$(\operatorname{coker}_{\text{pre}} \varphi)(U) = {}^{\mathscr{G}(U)}\big/_{\varphi(\mathscr{F}(U))}$$

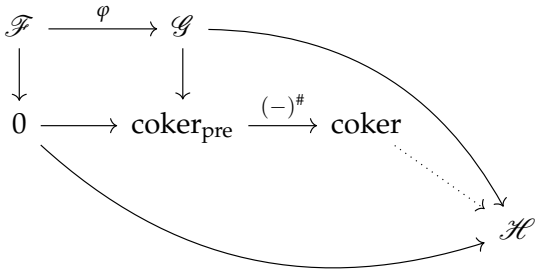
Furthermore for every $x \in X$ $\quad (\operatorname{coker} \varphi)_x = \operatorname{coker} \varphi_x$.

Prove that this is the cokernel in PSh.

Proof. Given a commutative diagram



We construct



□

Therefore we have proven that $\mathscr{O}_X$-Mod is an Abelian category and in particular by $\mathscr{O}_X$-Mod and sheaves of abelian groups $\mathsf{Sh}_{\mathsf{Ab}}(X)$ is an Abelian category.

10.4 Sequences in $\mathscr{O}_X$-Mod

Theorem 10.4.1. *The sequence of $\mathscr{O}_X$-modules $\mathscr{F} \to \mathscr{G} \to \mathscr{H}$ is exact at $\mathscr{G}$ if and only if for every $x \in X$ the sequence $\mathscr{F}_x \to \mathscr{G}_x \to \mathscr{H}_x$ is exact at $\mathscr{G}_x$.*

Probably not a *theorem* but a direct consequence of the commutatitivity of stalks and (co)kernels

Maybe talk about resolutions / types of resolutions here?

Appendix A

Algebra

Definition A.0.1 (Algebra)**.** ⌟

Definition A.0.2 (Local ring)**.** ⌟

Definition A.0.3 (Module)**.** ⌟

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